



SECP2523: DATABASE

SEM I 2024/2025

DATABASE DESIGN DESCRIPTIONS: LOGICAL

DATABASE DESIGN

FOR KADA COOPERATIVE ONLINE SYSTEM

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(KADA) Cooperative Berhad

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NOTE: A module can consist of one or multiple (related) use case. A team member in a group must be responsible to at least one module.

1. Introduction

1.1 Overview about the company

Kemubu Agricultural Development Authority (KADA) is a prominent agricultural development organization in Malaysia, established to support and develop the agricultural sector in the Kelantan region. Operating under the Ministry of Agriculture and Food Industries, KADA plays a crucial role in implementing agricultural development programs, managing irrigation systems, and supporting local farmers through cooperative initiatives.

The organization's cooperative division manages membership applications, coordinates agricultural support programs, and facilitates member services. Currently serving thousands of farmers across the Kelantan region, KADA's cooperative system is integral to providing financial assistance, technical support, and agricultural resources to its members.

1.2 The Existing System

The cooperation system used by KADA is currently being done manually. All the forms and paperwork are submitted and stored in physical condition by the client and the organization. Applicants need to attend the KADA office to obtain the application form, and there is no computerized system for the submission of the documents required. The application type is divided into two, member application and loan application. If a client wants to apply for a loan, one must first become a member of KADA cooperation. The applicant needs to submit a member application and pay RM50 to become a member. When the application for membership is approved, then only they can proceed with loan application. Applicants' information is kept in a physical paper and the clerk must check their application one by one. Then, board members of KADA will conduct a weekly meeting to verify the application and decide whether to accept or reject the application. Once the result is finalized, the applicant will be notified through SMS messages. KADA offers a various type of loan for its member including Al-Bai, Al-Innah, Vehicle Restoration, Road Tax & Insurance and Al-Qadrul Hassan. Applicants can choose what type of loan they want to apply for according to their needs. The processes for loan application are quite similar to member applications where applicants must come to the KADA office and obtain the application form, thus submitting it with all the documents related to the office physically. A meeting between board members will be conducted also to determine the decision regarding the application. Lastly, successful applicants will be notified through SMS messages.

In addition, KADA's cooperation system includes a shares system which all members are eligible to join. Members can contribute to the organization's capital through purchasing shares. These shares represent their ownership in the organization. Upon first time to apply for membership, the applicant needs to pay RM50. The registration fee will then be recorded as the applicant's shares when they are successful in becoming a member. Additionally, the shares amount will also put into factors which the member is eligible to apply for loan or not, all members must have at least RM350 worth of shares for them to apply for loan. If their shares do not fulfill the condition, they must first purchase the shares until they reach the required minimum amount.

a. **Issues with existing system**

The issues arising from the manual handling of the application are:

- i) **Data Security & Reliability** – All the related documents containing sensitive information about customers were stored physically. There might be risk of losing or leaking the data to malicious party because of human error. Moreover, the organization of the documents might not be in a systematic order, and it is possible for the data to mistakenly match with the wrong customer and end up providing misinformation about the customer. Thus, understanding the importance of having a convenient system in an organization might help the organization to provide secure and more reliable data for the customer.
- ii) **Timeliness** – The process of requiring and submitting documents is time consuming as the applicant needs to travel to KADA office multiple times. It requires a large amount of time just for the enquiries and submission of the documents. Some of the members might be living far away across the country and the distance between KADA office and their hometown must be put into consideration. To put it simply, the organization must provide a centralized system for the documentation which is accessible by the customer whenever or wherever they want.
- iii) **Scalability** – As time grows, KADA might experience an increasing number of its members. The manual system that is currently being used may struggle to handle a large volume of requests and applications in the future. Besides, the weekly meetings for applications approval might become a bottleneck and lead to slower response to the applicant, which will frustrate applicants. In summary, KADA needs to computerize their systems to give the best customer experience and enhance the efficiency of the system.

1.3 The Proposed System

The proposed cooperative system for KADA is a digital portal designed to simplify and streamline membership and loan applications processes. Applicants can apply for membership online, and once approved, they become members and gain access to loan services. Admins manage and verify applications, while Board Members (ALK) are responsible for making final member and loan approval decisions. The system will help to improve efficiency, provide secure data storage, and enhance the overall user experience.

The system will serve four types of users:

1. Applicants:

- Individuals applying for KADA Cooperation membership
- Applicants must submit an online application form and pay the required membership fee
- They receive real-time updates about their application status

2. Members:

- Once approved, applicants become members and can apply for loans anytime
- Members can use the loan calculator to estimate repayment plans and track their loan application statuses.

3. Admins:

- Admins handle all administrative tasks, including verifying membership and loan applications.
- They update applications statuses and forward approved loan applications to the Board Members (ALK) for final decisions.
- Admins also manage user data and ensure smooth operation of the system.

4. Board Members (ALK):

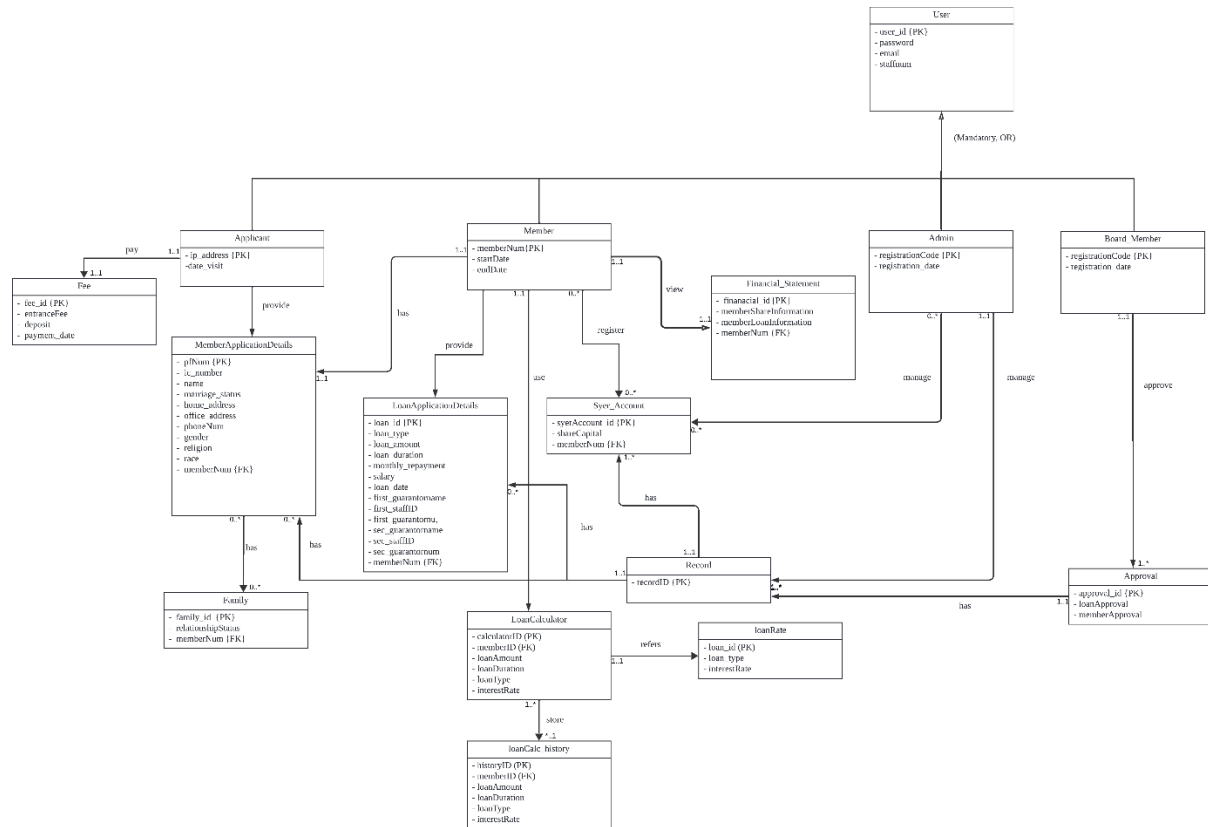
- Board Members review loan applications forwarded by Admins.
- They make the final decision on loan approvals or rejections based on eligibility criteria and KADA policies.

Key Features:

- Online membership registration for applicants.
- Loan application and tracking for members.
- Real-time updates on application status via email notifications.
- A loan calculator to help members plan repayment schedules.
- A centralized system for admins and board members to manage and review applications.

2. Global Conceptual Data Model (Global ERD)

2.1 Global Conceptual Database Design (ERD)



2.2 Data Dictionary for the Global Conceptual ERD

1. User

FIELD NAME	DATA TYPE	FIELD SIZE	DESCRIPTION	EXAMPLE
user_id	INT	11	Unique identifier for each user. (PK)	1
password	VARCHAR	255	User's password (should be hashed for security).	*Hashed value*
email	VARCHAR	255	User's email address.	[email address removed]

staffnum	VARCH AR	20	User's staff number (if applicable, could be a unique identifier for employees).	S1234
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2. Member

FIELD NAME	DATA TYPE	FIELD SIZE	DESCRIPTION	EXAMPLE
member Num	VARCH AR	20	Unique identifier for each member.	M20230001
startDate	DATE		Membership start date.	2023-12-01
endDate	DATE		Membership end date.	2024-12-01

3.Admin

FIELD NAME	DATA TYPE	FIELD SIZE	DESCRIPTION	EXAMPLE
registrati onCode	VARC HAR	20	Unique registration code for each administrator.	ADMIN-0005
registrati on_date	DATE		Administrator registration date.	2023-09-15

4.Board_Member

FIELD NAME	DATA TYPE	FIELD SIZE	DESCRIPTION	EXAMPLE
registrati onCode	VARC HAR	20	Unique registration code for each board member.	BOARD-0012
registrati on_date	DATE		Board member registration date.	2023-08-20

5. Fee

FIELD NAME	DATA TYPE	FIELD SIZE	DESCRIPTION	EXAMPLE
fee_id	INT	11	Unique identifier for each fee record. (PK)	1
entranceFee	DECIMAL	11	Amount of the entrance fee.	500
deposit	DECIMAL	11	Amount of the deposit.	200
payment_date	DATE		Date when the fee payment was made.	2023-10-27

6. MemberApplicationDetails

FIELD NAME	DATA TYPE	FIELD SIZE	DESCRIPTION	EXAMPLE
pfNum	VARCHAR	20	Unique identifier for each member.	M20230001
name	VARCHAR	255	Member's full name.	Siti Nurhaliza
ic_number	VARCHAR	14	Member's identity card number (Malaysian IC format).	901201-10-5678
marriage_status	VARCHAR	20	Member's marital status (e.g., Single, Married, Divorced).	Married
gender	VARCHAR	10	Member's gender.	Male
religion	VARCHAR	20	Member's religion.	Islam
race	VARCHAR	20	Member's race.	Malay
phoneNum	VARCHAR	20	Member's phone number.	012-3456789
memberNum	INT	11	Foreign key referencing Member.memberNum	1001

7.LoanApplicationDetails

FIELD NAME	DATA TYPE	FIELD SIZE	DESCRIPTION	EXAMPLE
loan_id	INT	11	Unique identifier for each loan application. (PK)	501
loan_type	VARCHAR	50	Type of loan (e.g., Al-Bai, Al-Innah, etc.).	Al-Bai
loan_amount	DECIMAL	11	The total amount of the loan.	10000
loan_duration	INT	4	Duration of the loan (in months or years).	24
monthly_repayment	DECIMAL	11	The calculated monthly repayment amount.	458.33
salary	DECIMAL	11	Applicant's salary.	6000
loan_date	DATE		The date the loan application was submitted.	2023-10-27
first_guarantorname	VARCHAR	255	Name of the first guarantor.	Ahmad Ali
first_staffID	VARCHAR	20	Staff ID of the first guarantor (if applicable).	S1234
first_guarantornum	VARCHAR	20	Contact number of the first guarantor.	012-3456789
sec_guarantorname	VARCHAR	255	Name of the second guarantor.	Siti Rahman
sec_staffID	VARCHAR	20	Staff ID of the second guarantor (if applicable).	S5678
sec_guarantornum	VARCHAR	20	Contact number of the second guarantor.	017-9876543
memberNum	VARCHAR	20	Foreign key referencing the Member table, indicating the member associated with this loan application. (FK)	M20230001

8.Syer_Account

FIELD NAME	DATA TYPE	FIELD SIZE	DESCRIPTION	EXAMPLE
syerAccountID	VARCHAR	20	Unique identifier for each item account.	001
shareCapital	DECIMAL	11	Share capital amount.	10000
memberNum	VARCHAR	20	Foreign key referencing Member.memberNum.	M20230001

9. Financial_Statement

FIELD NAME	DATA TYPE	FIELD SIZE	DESCRIPTION	EXAMPLE
financial_id	INT	11	Unique identifier for each financial statement record. (PK)	12
memberShareInformation	VARCHAR	255	Information about member shares	Share details
memberLoanInformation	VARCHAR	255	Information about member loans	Loan details

10. Family

FIELD NAME	DATA TYPE	FIELD SIZE	DESCRIPTION	EXAMPLE
familyID	INT	11	Unique identifier for each family.	201
relationshipStatus	VARCHAR	50	Relationship to the member.	Spouse
memberNum	VARCHAR	20	Foreign key referencing Member.memberNum.	M20230001

11.LoanCalculator

FIELD NAME	DATA TYPE	FIELD SIZE	DESCRIPTION	EXAMPLE
calculation_id	INT	11	Unique identifier for each loan calculation. (PK)	301
memberID	VARCHAR	20	Foreign key referencing the Member table, indicating the member associated with this calculation. (FK)	M20230001
loanAmount	DECIMAL	11	The total amount of the loan being calculated.	5000
loanDuration	INT	4	Duration of the loan (in months or years).	24
loanType	VARCHAR	50	Type of loan (e.g., Al-Bai, Al-Innah, etc.).	Al-Innah
interestRate	DECIMAL	5	Interest rate used in the calculation (e.g., 5.25 for 5.25%).	7.5

12.Applicant

FIELD NAME	DATA TYPE	FIELD SIZE	DESCRIPTION	EXAMPLE
ip_address	VARCHAR	255	Applicant's IP address and address combined. (PK)	901201-10-5678, 123 Jalan Utama
startDate	DATE		Application start date.	2023-10-26
endDate	DATE		Application end date.	2023-11-15

13. LoanCalc_History

FIELD NAME	DATA TYPE	FIELD SIZE	DESCRIPTION	EXAMPLE
historyID	INT	11	Unique identifier for each loan calculation history record. (PK)	701
memberID	VARCHAR	20	Foreign key referencing the Member table, indicating the	M20230001

			member associated with this historical calculation. (FK)	
loanAmount	DECIMAL	11	The total amount of the loan	5000
loanDuration	INT	4	Duration of the loan (in months or years) in the historical calculation.	24
loanType	VARCHAR	50	Type of loan (e.g., Al-Bai, Al-Innah, etc.) in the historical calculation.	Al-Innah
interestRate	DECIMAL	5	Interest rate used in the historical calculation (e.g., 5.25 for 5.25%).	7.5

14.Record

FIELD NAME	DATA TYPE	FIELD SIZE	DESCRIPTION	EXAMPLE
record_id	INT	11	Unique identifier for each installment record. (PK)	500

15.Member_Savings

FIELD NAME	DATA TYPE	FIELD SIZE	DESCRIPTION	EXAMPLE
Savings_id	INT	11	Unique identifier for saving records. (PK)	
memberNum	INT	20	Links to the member owning this savings.	
month	DATE	255	The month the savings were made.	
amount	DECIMAL	20	Amount saved by the member.	

16. Shares

FIELD NAME	DATA TYPE	FIELD SIZE	DESCRIPTION	EXAMPLE
share_id	INT		Unique identifier for shares records.(PK)	
memberNum	INT		Links to the member owning these shares.(FK)	
total_shares	DECIMAL		Total shares owned by the member.	

17. Dividend

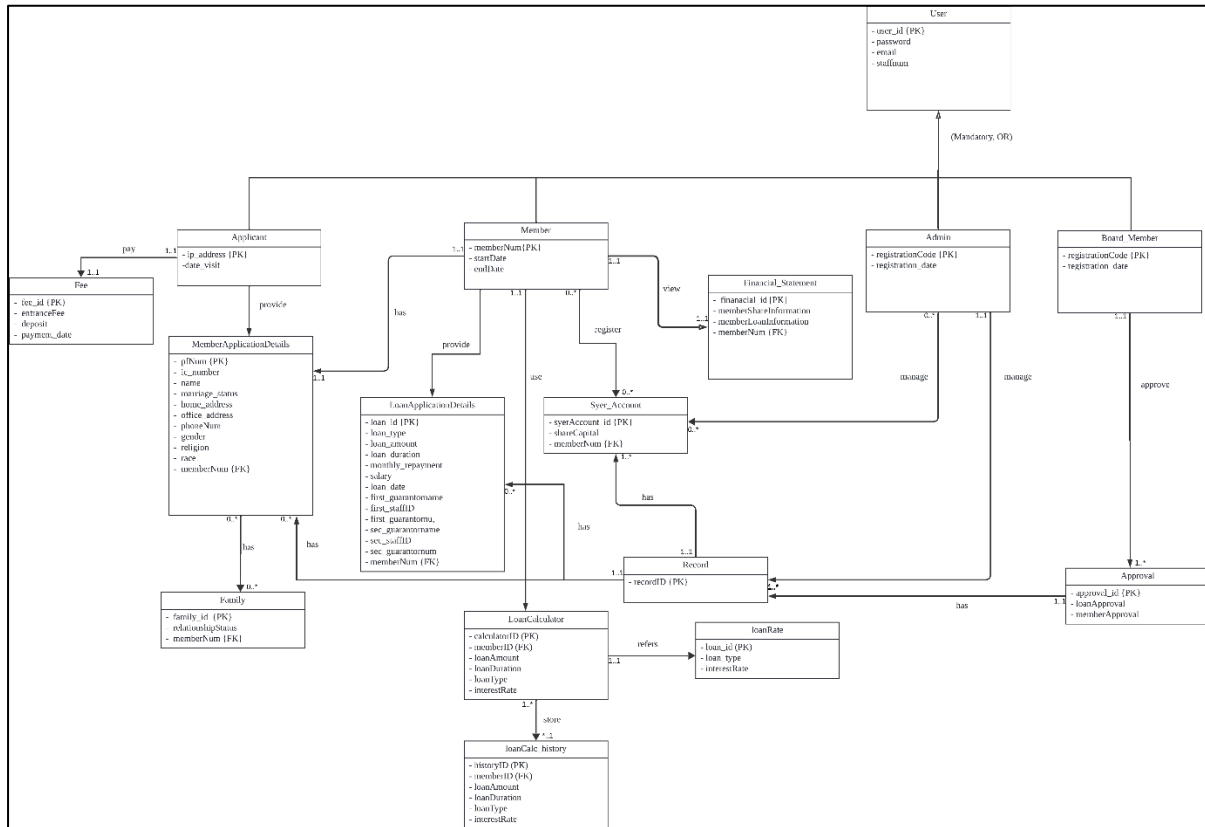
FIELD NAME	DATA TYPE	FIELD SIZE	DESCRIPTION	EXAMPLE
dividend_id	INT		Unique identifier for dividend records.(PK)	
year	INT		The year for the dividend.	
rate	DECIMAL		Dividend rate for the year.	

18. Reports

FIELD NAME	DATA TYPE	FIELD SIZE	DESCRIPTION	EXAMPLE
report_id	INT		Unique identifier for report records.(PK)	
memberNum	INT		Links to the member associated with the report.(FK)	
report_date	DATE		The date the report was generated.	
monthly_savings	DECIMAL		Total monthly savings for the member.	
yearly_profit	DECIMAL		Calculated yearly profit for the member.	

3. Logical Data Model (ERD)

3.1 Logical Database Design (Logical ERD)



3.2 Data Dictionary for the Logical ERD

1. User

FIELD NAME	DATA TYPE	FIELD SIZE	DESCRIPTION	EXAMPLE
user_id	INT	11	Unique identifier for each user. (PK)	1
password	VARCHAR	255	User's password (should be hashed for security).	*Hashed value*
email	VARCHAR	255	User's email address.	[email address removed]
staffnum	VARCHAR	20	User's staff number (if applicable, could be a unique identifier for employees).	S1234

2. Member

FIELD NAME	DATA TYPE	FIELD SIZE	DESCRIPTION	EXAMPLE
memberNum	VARCHAR	20	Unique identifier for each member.	M20230001
startDate	DATE		Membership start date.	2023-12-01
endDate	DATE		Membership end date.	2024-12-01

3. Admin

FIELD NAME	DATA TYPE	FIELD SIZE	DESCRIPTION	EXAMPLE
registrationCode	VARCHAR	20	Unique registration code for each administrator.	ADMIN-0005
registration_date	DATE		Administrator registration date.	2023-09-15

4. Board_Member

FIELD NAME	DATA TYPE	FIELD SIZE	DESCRIPTION	EXAMPLE
registrationCode	VARCHAR	20	Unique registration code for each board member.	BOARD-0012
registration_date	DATE		Board member registration date.	2023-08-20

5. Fee

FIELD NAME	DATA TYPE	FIELD SIZE	DESCRIPTION	EXAMPLE
fee_id	INT	11	Unique identifier for each fee record. (PK)	1
entranceFee	DECIMAL	11	Amount of the entrance fee.	500
deposit	DECIMAL	11	Amount of the deposit.	200
payment_date	DATE		Date when the fee payment was made.	2023-10-27

6. MemberApplicationDetails

FIELD NAME	DATA TYPE	FIELD SIZE	DESCRIPTION	EXAMPLE
pfNum	VARCHAR	20	Unique identifier for each member.	M20230001
name	VARCHAR	255	Member's full name.	Siti Nurhaliza
ic_number	VARCHAR	14	Member's identity card number (Malaysian IC format).	901201-10-5678
marriage_status	VARCHAR	20	Member's marital status (e.g., Single, Married, Divorced).	Married
gender	VARCHAR	10	Member's gender.	Male
religion	VARCHAR	20	Member's religion.	Islam
race	VARCHAR	20	Member's race.	Malay
phoneNum	VARCHAR	20	Member's phone number.	012-3456789

memberNum	INT	11	Foreign key referencing Member.memberNum	1001
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7.LoanApplicationDetails

FIELD NAME	DATA TYPE	FIELD SIZE	DESCRIPTION	EXAMPLE
loan_id	INT	11	Unique identifier for each loan application. (PK)	501
loan_type	VARCHAR	50	Type of loan (e.g., Al-Bai, Al-Innah, etc.).	Al-Bai
loan_amount	DECIMAL	11	The total amount of the loan.	10000
loan_duration	INT	4	Duration of the loan (in months or years).	24
monthly_repayment	DECIMAL	11	The calculated monthly repayment amount.	458.33
salary	DECIMAL	11	Applicant's salary.	6000
loan_date	DATE		The date the loan application was submitted.	2023-10-27
first_guarantortname	VARCHAR	255	Name of the first guarantor.	Ahmad Ali
first_staffID	VARCHAR	20	Staff ID of the first guarantor (if applicable).	S1234
first_guarantortnum	VARCHAR	20	Contact number of the first guarantor.	012-3456789
sec_guarantortname	VARCHAR	255	Name of the second guarantor.	Siti Rahman
sec_staffID	VARCHAR	20	Staff ID of the second guarantor (if applicable).	S5678
sec_guarantortnum	VARCHAR	20	Contact number of the second guarantor.	017-9876543
memberNum	VARCHAR	20	Foreign key referencing the Member table, indicating the member associated with this loan application. (FK)	M20230001

8.Saving_Account

FIELD NAME	DATA TYPE	FIELD SIZE	DESCRIPTION	EXAMPLE
syerAccountID	VARCHAR	20	Unique identifier for each item account.	001
shareCapital	DECIMAL	11	Share capital amount.	10000
memberNum	VARCHAR	20	Foreign key referencing Member.memberNum.	M20230001

9. Financial_Statement

FIELD NAME	DATA TYPE	FIELD SIZE	DESCRIPTION	EXAMPLE
financial_id	INT	11	Unique identifier for each financial statement record. (PK)	12
memberShareInformation	VARCHAR	255	Information about member shares	Share details
memberLoanInformation	VARCHAR	255	Information about member loans	Loan details

10. Family

FIELD NAME	DATA TYPE	FIELD SIZE	DESCRIPTION	EXAMPLE
familyID	INT	11	Unique identifier for each family.	201
relationshipStatus	VARCHAR	50	Relationship to the member.	Spouse
memberNum	VARCHAR	20	Foreign key referencing Member.memberNum.	M20230001

11.LoanCalculator

FIELD NAME	DATA TYPE	FIELD SIZE	DESCRIPTION	EXAMPLE
calculation_id	INT	11	Unique identifier for each loan calculation. (PK)	301
memberID	VARCHAR	20	Foreign key referencing the Member table, indicating the member associated with this calculation. (FK)	M20230001
loanAmount	DECIMAL	11	The total amount of the loan being calculated.	5000
loanDuration	INT	4	Duration of the loan (in months or years).	24
loanType	VARCHAR	50	Type of loan (e.g., Al-Bai, Al-Innah, etc.).	Al-Innah
interestRate	DECIMAL	5	Interest rate used in the calculation (e.g., 5.25 for 5.25%).	7.5

12.Applicant

FIELD NAME	DATA TYPE	FIELD SIZE	DESCRIPTION	EXAMPLE
ip_address	VARCHAR	255	Applicant's IP address and address combined. (PK)	901201-10-5678, 123 Jalan Utama
startDate	DATE		Application start date.	2023-10-26
endDate	DATE		Application end date.	2023-11-15

13. LoanCalc_History

FIELD NAME	DATA TYPE	FIELD SIZE	DESCRIPTION	EXAMPLE
historyID	INT	11	Unique identifier for each loan calculation history record. (PK)	701
memberID	VARCHAR	20	Foreign key referencing the Member table, indicating the member associated with this historical calculation. (FK)	M20230001
loanAmount	DECIMAL	11	The total amount of the loan	5000
loanDuration	INT	4	Duration of the loan (in months or years) in the historical calculation.	24
loanType	VARCHAR	50	Type of loan (e.g., Al-Bai, Al-Innah, etc.) in the historical calculation.	Al-Innah
interestRate	DECIMAL	5	Interest rate used in the historical calculation (e.g., 5.25 for 5.25%).	7.5

14.Record

FIELD NAME	DATA TYPE	FIELD SIZE	DESCRIPTION	EXAMPLE
record_id	INT	11	Unique identifier for each installment record. (PK)	500

15.Member_Savings

FIELD NAME	DATA TYPE	FIELD SIZE	DESCRIPTION	EXAMPLE
Savings_id	INT	11	Unique identifier for saving records. (PK)	
memberNum	INT	20	Links to the member owning this savings.	
month	DATE	255	The month the savings were made.	
amount	DECIMAL	20	Amount saved by the member.	

16. Shares

FIELD NAME	DATA TYPE	FIELD SIZE	DESCRIPTION	EXAMPLE
share_id	INT		Unique identifier for shares records.(PK)	
memberNum	INT		Links to the member owning these shares.(FK)	
total_shares	DECIMAL		Total shares owned by the member.	

17. Dividend

FIELD NAME	DATA TYPE	FIELD SIZE	DESCRIPTION	EXAMPLE
dividend_id	INT		Unique identifier for dividend records.(PK)	
year	INT		The year for the dividend.	
rate	DECIMAL		Dividend rate for the year.	

18. Reports

FIELD NAME	DATA TYPE	FIELD SIZE	DESCRIPTION	EXAMPLE
report_id	INT		Unique identifier for report records.(PK)	
memberNum	INT		Links to the member associated with the report.(FK)	
report_date	DATE		The date the report was generated.	
monthly_savings	DECIMAL		Total monthly savings for the member.	
yearly_profit	DECIMAL		Calculated yearly profit for the member.	

3.3 Relational Database Schema(s)

3.3.1 User

User(user_id, password, email, staffnum)

3.3.2 Member

Member(memberNum, startDate, endDate)

3.3.3 Admin

Admin(registrationCode, registrationDate)

3.3.4 Board_Member

Board_Member(registrationCode, registrationDate)

3.3.5 Fee

Fee(fee_id, entranceFee, deposit, paymentDate)

3.3.6 MemberApplicationDetails

MemberApplicationDetails(pfNum, name, ic_number, marriage_status, gender, religion, race, phoneNum, memberNum)

3.3.7 LoanApplicationDetails

LoanApplicationDetails(loan_id, loan_type, loan_amount, loan_duration, monthly_repayment, salary, loan_date, first_guarantor_name, first_staff_ID, first_guarantor_contact, second_guarantor_name, second_staff_ID, second_guarantor_contact, memberNum)

3.3.8 Syer_Account

Syer_Account(syerAccountID, shareCapital, memberNum)

3.3.9 Financial_Statement

Financial_Statement(financial_id, memberShareInformation, memberLoanInformation)

3.3.10 Family

Family(familyID, relationshipStatus, memberNum)

3.3.11 LoanCalculator

LoanCalculator(calculation_id, memberID, loanAmount, loanDuration, loanType, interestRate)

3.3.12 Applicant

Applicant(ip_address, startDate, endDate)

3.3.13 LoanCalc_History

LoanCalc_History(historyID, memberID, loanAmount, loanDuration, loanType, interestRate)

3.3.14 Record

Record(record_id)

3.3.15 Member_Savings

Member_Savings(Savings_id, memberNum, month, amount)

3.3.16 Shares

Shares(share_id, memberNum, total_shares)

3.3.17 Dividend

Dividend(dividend_id, year, rate)

3.3.18 Reports

Reports(report_id, memberNum, report_date, monthly_savings, yearly_profit)

3.4 Normalization

3.4.1 User

User(user_id, password, email, staffnum)

Fd1: user_id $\rightarrow$ password, email, staffnum

1NF & 2NF & 3NF & BCNF:

USER(user_id, password, email, staffnum)

3.4.2 Member

Member(memberNum, startDate, endDate)

Fd1: memberNum $\rightarrow$ startDate, endDate

1NF & 2NF & 3NF & BCNF:

Member(memberNum, startDate, endDate)

3.4.3 Admin

Admin(registrationCode, registrationDate)

Fd1: registrationCode $\rightarrow$ registrationDate

1NF & 2NF & 3NF & BCNF:

Admin(registrationCode, registrationDate)

3.4.4 Board_Member

Board_Member(registrationCode, registrationDate)

Fd1: registrationCode $\rightarrow$ registrationDate

1NF & 2NF & 3NF & BCNF:

Board_Member(registrationCode, registrationDate)

3.4.5 Fee

Fee(fee_id, entranceFee, deposit, paymentDate)

Fd1: fee_id $\rightarrow$ entranceFee, deposit, paymentDate

1NF & 2NF & 3NF & BCNF:

Fee(fee_id, entranceFee, deposit, paymentDate)

3.4.6 MemberApplicationDetails

MemberApplicationDetails(pfNum, name, ic_number, marriage_status, gender, religion, race, phoneNum, memberNum)

Fd1: pfNum → name, ic_number, marriage_status, gender, religion, race, phoneNum, memberNum

1NF & 2NF & 3NF & BCNF:

MemberApplicationDetails(pfNum, name, ic_number, marriage_status, gender, religion, race, phoneNum, memberNum)

3.4.7 LoanApplicationDetails

LoanApplicationDetails(loan_id, loan_type, loan_amount, loan_duration, monthly_repayment, salary, loan_date, first_guarantor_name, first_staff_ID, first_guarantor_contact, second_guarantor_name, second_staff_ID, second_guarantor_contact, memberNum)

Fd1: loan_id → loan_type, loan_amount, loan_duration, monthly_repayment, salary, loan_date, first_guarantor_name, first_staff_ID, first_guarantor_contact, second_guarantor_name, second_staff_ID, second_guarantor_contact, memberNum

1NF & 2NF & 3NF & BCNF:

LoanApplicationDetails(loan_id, loan_type, loan_amount, loan_duration, monthly_repayment, salary, loan_date, first_guarantor_name, first_staff_ID, first_guarantor_contact, second_guarantor_name, second_staff_ID, second_guarantor_contact, memberNum)

3.4.8 Syer_Account

Syer_Account(syerAccountID, shareCapital, memberNum)

Fd1: syerAccountID → shareCapital, memberNum

1NF & 2NF & 3NF & BCNF:

Syer_Account(syerAccountID, shareCapital, memberNum)

3.4.9 Financial_Statement

Financial_Statement(financial_id, memberShareInformation, memberLoanInformation)

Fd1: financial_id → memberShareInformation, memberLoanInformation

1NF & 2NF & 3NF & BCNF:

Financial_Statement(financial_id, memberShareInformation, memberLoanInformation)

3.4.10 Family

Family(familyID, relationshipStatus, memberNum)

Fd1: familyID $\rightarrow$ relationshipStatus, memberNum

1NF & 2NF & 3NF & BCNF:

Family(familyID, relationshipStatus, memberNum)

3.4.11 LoanCalculator

LoanCalculator(calculation_id, memberID, loanAmount, loanDuration, loanType, interestRate)

Fd1: calculation_id $\rightarrow$ memberID, loanAmount, loanDuration, loanType, interestRate

1NF & 2NF & 3NF & BCNF:

LoanCalculator(calculation_id, memberID, loanAmount, loanDuration, loanType, interestRate)

3.4.12 Applicant

Applicant(ip_address, startDate, endDate)

Fd1: ip_address $\rightarrow$ startDate, endDate

1NF & 2NF & 3NF & BCNF:

Applicant(ip_address, startDate, endDate)

3.4.13 LoanCalc_History

LoanCalc_History(historyID, memberID, loanAmount, loanDuration, loanType, interestRate)

Fd1: historyID $\rightarrow$ memberID, loanAmount, loanDuration, loanType, interestRate

1NF & 2NF & 3NF & BCNF:

LoanCalc_History(historyID, memberID, loanAmount, loanDuration, loanType, interestRate)

3.4.14 Record

Record(record_id)

Fd1: record_id $\rightarrow$ (no other attributes)

(This table has only one column shown. If it truly has no non-key attributes, it's already in BCNF.)

1NF & 2NF & 3NF & BCNF:

Record(record_id)

3.4.15 Member_Savings

Member_Savings(Savings_id, memberNum, month, amount)

Fd1: Savings_id $\rightarrow$ memberNum, month, amount

1NF & 2NF & 3NF & BCNF:

Member_Savings(Savings_id, memberNum, month, amount)

3.4.16 Shares

Shares(share_id, memberNum, total_shares)

Fd1: share_id $\rightarrow$ memberNum, total_shares

1NF & 2NF & 3NF & BCNF:

Shares(share_id, memberNum, total_shares)

3.4.17 Dividend

Dividend(dividend_id, year, rate)

Fd1: dividend_id $\rightarrow$ year, rate

1NF & 2NF & 3NF & BCNF:

Dividend(dividend_id, year, rate)

3.4.18 Reports

Reports(report_id, memberNum, report_date, monthly_savings, yearly_profit)

Fd1: report_id $\rightarrow$ memberNum, report_date, monthly_savings, yearly_profit

1NF & 2NF & 3NF & BCNF:

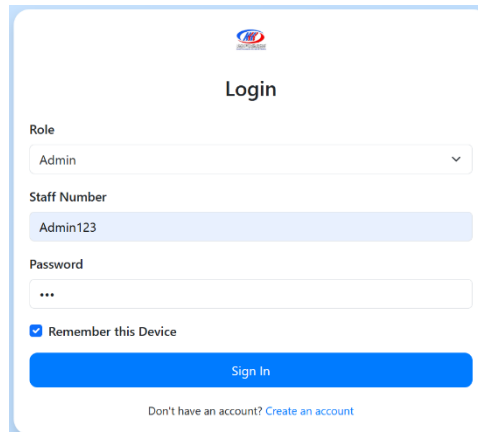
Reports(report_id, memberNum, report_date, monthly_savings, yearly_profit)

4. Interface Design and SQL Implementation

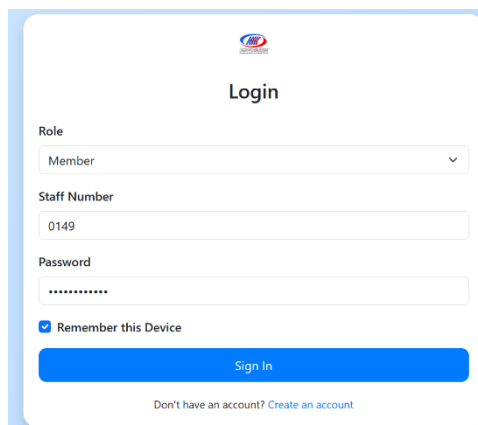
4.1 Interface Design (Mapped to the Proposed SQL Statement)

4.1.1 User Authentication

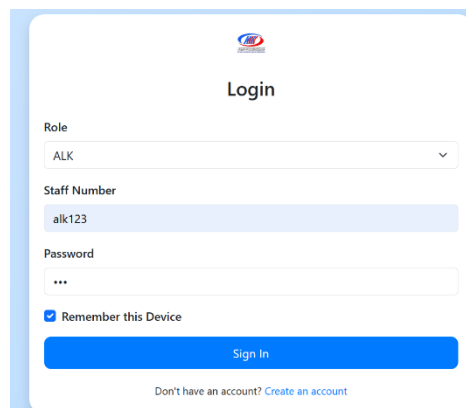
Login to respective page (Member, Admin, ALK)



The login form for the Admin role features the JNU logo at the top. Below the title "Login", there is a dropdown menu for "Role" set to "Admin". The "Staff Number" field contains "Admin123". The "Password" field is masked with three dots. A checked checkbox labeled "Remember this Device" is present. A blue "Sign In" button is at the bottom, with a link "Don't have an account? Create an account" below it.



The login form for the Member role features the JNU logo at the top. Below the title "Login", there is a dropdown menu for "Role" set to "Member". The "Staff Number" field contains "0149". The "Password" field is masked with eight dots. A checked checkbox labeled "Remember this Device" is present. A blue "Sign In" button is at the bottom, with a link "Don't have an account? Create an account" below it.



The login form for the ALK role features the JNU logo at the top. Below the title "Login", there is a dropdown menu for "Role" set to "ALK". The "Staff Number" field contains "alk123". The "Password" field is masked with three dots. A checked checkbox labeled "Remember this Device" is present. A blue "Sign In" button is at the bottom, with a link "Don't have an account? Create an account" below it.

4.1.2 Member Application

Insert value into membership application form – Personal Details

Permohonan Keahlian - Maklumat Peribadi

Nama

No KP

Emel

Jantina

Agama

Bangsa

Alamat Kediaman

No Poskod

Negeri

No. Tel Bimbit

No. Tel Rumah

Nombor Anggota

Nombor PF

Gaji Bulanan

Seterusnya

Insert value into membership application form – Family Details

Permohonan Keahlian - Maklumat Keluarga

Nama Pewaris

Hubungan Pewaris

No. KP Pewaris

Sebelum **Seterusnya**

Insert value into membership application form – Payment Details

Permohonan Keahlian - Pembayaran

Yuran Masuk RM 50

Muat Naik Resit Bayaran

Choose File receipt.png

Sebelum **Hantar Permohonan**

4.1.3 Loan Application

Insert value into Loan application form – Loan Application Details

Successful Application Indication through email notification

4.1.4 Savings Application

Insert Value for Deposit Saving Account inside the form (All Share Fee, Pay month)

The screenshot shows the 'Maklumat Pembayaran' form in the 'Akaun Simpanan' system. The form is titled 'Maklumat Pembayaran' and includes a sub-header 'Sila masukkan butiran Maklumat Pembayaran anda'. It contains several input fields for payment details:

- Modal Syer (Minimum RM: 50.00): MYR 50
- Modal Yuran (Minimum RM: 35.00): MYR 35
- Sumbangan Tabung Kebajikan: MYR 5
- Simpanan Tetap: MYR 5
- Simpanan Anggota: MYR 20
- Bulan Pembayaran: Bulan Jun

A blue button labeled 'Teruskan ke Pembayaran' is located at the bottom of the form.

Insert payment details and Check summary of payment inside payment summary

The screenshot shows the 'Semakan Pembayaran' (Payment Summary) page in the 'Akaun Simpanan' system. The page is titled 'Semakan Pembayaran' and includes a sub-header 'Wang anda akan diproses dan dipindahkan ke akaun simpanan anda dalam masa 1-2 hari bekerja selepas kami menerima dana tersebut.' It contains a table of payment details and a summary of the payment.

Butiran Pemindahan	
Modal Syer	50.00
Modal Yuran	35.00
Simpanan Tetap	5.00
Tabung Anggota	5.00
Simpanan Anggota	20.00
Bayaran Pinjaman	Tiada bayaran pinjaman.
Akaun	Akaun Koperasi KADA

Kaedah Pembayaran
Kaedah Pembayaran: Pilih kaedah
Nombor Staff: Masukkan nombor staff anda
Sila masukkan nombor staff anda

Rumusan Pembayaran

Yuran Transaksi	MYR 0.00
Masa Pemrosesan	1-2 hari bekerja
Jumlah Keseluruhan	MYR 115.00

A blue button labeled 'Teruskan' is located at the bottom of the page.

Update the transaction status for the deposit

 **Semakan transaksi**

 Laman Utama

 Semakan Permohonan

 Semakan Transaksi

 Senarai Permohonan Diterima

 Laporan Kerani

 Log Keluar



Transaksi Simpanan Tertunda
Transaksi akan diproses dalam 1-2 hari bekerja selepas kelulusan

No. Kakitangan 1443	Modal Syer 50.00	Modal Yuran 35.00
Sumbangan Tabung Kebajikan 5.00	Simpanan Tetap 5.00	Simpanan Anggota 20.00

Lulus

Tolak

Transaksi Pinjaman Tertunda
Transaksi akan diproses dalam 1-2 hari bekerja selepas kelulusan

Tiada Transaksi Pinjaman Tertunda.
Pada masa ini tiada transaksi pinjaman tertunda.

 Pentadbir 1

4.1.5 Review Applications

Query the applications that are not reviewed yet (membership/loan)

Semakan Permohonan

Menu: Laman Utama, Semakan Permohonan, Semakan Transaksi, Senarai Permohonan Diterima, Laporan Kerani, Log Keluar

Permohonan Keahlian

Nama	Alamat Kediaman	No. Tel	Status Semakan	Tindakan	Perincian
Syahmi Faris bin Rusli	No 14, Taman Karak, Jalan Utama Karak	0198824509	Not Reviewed	Tanda Sebagai Disemak	Lihat Rincian

Permohonan Pinjaman

Nama	No. KP	Jenis Pinjaman	Amaun Pinjaman	Durasi (Tahun)	Status Semakan	Tindakan	Perincian
Tiada permohonan yang tersedia.							

Update the review status of the applications

Semakan Permohonan

Menu: Laman Utama, Semakan Permohonan, Semakan Transaksi, Senarai Permohonan Diterima, Laporan Kerani, Log Keluar

Permohonan Keahlian

Nama	Alamat Kediaman	No. Tel	Status Semakan	Tindakan	Perincian
Baidurah binti Ahmad Jamal			Reviewed	Tanda Sebagai Disemak	Lihat Rincian

Permohonan Pinjaman

Nama	No. KP	Jenis Pinjaman	Amaun Pinjaman	Durasi (Tahun)	Status Semakan	Tindakan	Perincian
Tiada permohonan yang tersedia.							

Mengesahkan Tindakan

Adakah anda sudah selesai menyemak permohonan ini?

Batal Hantar ke ALK

Query the pending applications (membership/loan)

 Dashboard

Laman Utama

Semakan Permohonan

Senarai Ahli

Log Keluar

Menu icon

ALK 1

Permohonan Keahlian

Nama	Alamat	No. Telefon	Status	Tindakan	Maklumat Lanjut
Syahmi Faris bin Rusli	No 14, Taman Karak, Jalan Utama Karak	0198824509	Dalam proses	TerimaTolak	Lihat

Permohonan Pinjaman

Nama	No. IC	Jenis Pinjaman	Jumlah Pinjaman	Tempoh (Tahun)	Status	Tindakan	Maklumat Lanjut
Tiada permohonan masuk.							

Update the status of the applications

 Dashboard

Laman Utama

Semakan Permohonan

Senarai Ahli

Log Keluar

Menu icon

ALK 1

Permohonan Keahlian

Nama	Alamat	No. Telefon	Status	Tindakan	Maklumat Lanjut
Baidurah binti Ahmad Jams			Dalam proses	TerimaTolak	Lihat

Permohonan Pinjaman

Nama	No. IC	Jenis Pinjaman	Jumlah Pinjaman	Tempoh (Tahun)	Status	Tindakan	Maklumat Lanjut
Tiada permohonan masuk.							

Konfirmasi Tindakan

Adakah awak ingin terima permohonan ini?

Batal

Setuju

4.1.6 Loan Calculator

Insert the data

Loan Calculator

Jumlah Pinjaman (RM):

Tempoh Pinjaman (Years):

Jenis Pinjaman:

Kadar Faedah (%):

Kira

Calculate based on the inserted data

Jawapan

Bayaran Bulanan: RM 267.72

Jumlah Bayaran Balik: RM 9,638.00

Close

4.1.7 Admin Report

Insert Member Data

Insert Member

Member Number

Start Date

End Date

Update Member Data

Update Member

Member Number

New Start Date

New End Date

Delete Member Data

Delete Member

Member Number

4.2 SQL Statements (DDL & DML)

4.2.1 User Authentication

DDL:

(a) Create accounts table

```
CREATE TABLE accounts (  
    id INT AUTO_INCREMENT PRIMARY KEY,  
    email VARCHAR(255) DEFAULT NULL,  
    password VARCHAR(255) DEFAULT NULL,  
    staffNum VARCHAR(255) DEFAULT NULL,  
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP  
);
```

DML:

(b) Login As Member, Admin, ALK

```
INSERT INTO accounts (  
    Role, staffNum, password  
)VALUES(  
    'Admin', 'Admin123', '123'  
);  
INSERT INTO accounts (  
    Role, staffNum, password  
)VALUES(  
    'ALK', 'alk123', '123'  
);  
INSERT INTO accounts (  
    Role, staffNum, password  
)VALUES(  
    'Member', '0149', '06215416024565'  
);
```

4.2.2 Member Application

DDL:

(a) Create member_applications table

```
CREATE TABLE member_applications (  
    id INT AUTO_INCREMENT PRIMARY KEY,  
    nama VARCHAR(100) NULL,  
    no_kp VARCHAR(20) NULL,  
    jantina VARCHAR(10) NULL,  
    agama VARCHAR(50) NULL,  
    bangsa VARCHAR(50) NULL,  
    alamat_rumah TEXT NULL,  
    poskod VARCHAR(10) NULL,  
    negeri VARCHAR(50) NULL,  
    no_tel_bimbit VARCHAR(20) NULL,  
    no_tel_rumah VARCHAR(20) NULL,  
    gaji_bulanan DECIMAL(10,2) NULL,  
    nama_pewaris VARCHAR(100) NULL,  
    hubungan_pewaris VARCHAR(50) NULL,  
    no_kp_pewaris VARCHAR(20) NULL,  
    fee_masuk DECIMAL(10,2) NULL,  
    receipt_path VARCHAR(255) NULL,  
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,  
    status VARCHAR(255) NOT NULL DEFAULT 'pending',  
    review_status ENUM('reviewed', 'not_reviewed') NOT NULL DEFAULT 'not_reviewed',  
    submission_date DATETIME DEFAULT CURRENT_TIMESTAMP,  
  
    staff_no VARCHAR(100) NULL,  
    email VARCHAR(255) NULL,  
    doneEmail ENUM('0', '1') NOT NULL DEFAULT '0',  
    no_pf VARCHAR(100) NULL  
);
```

DML:

(a) Insert value into member application – Personal Details

```
INSERT INTO member_applications (  
    nama, no_kp, email, jantina, agama, bangsa, alamat_rumah, poskod, negeri, no_tel_bimbit,  
    no_tel_rumah, gaji_bulanan, staff_no, no_pf  
) VALUES (  
    'Syahmi Faris', '2364727642', 'syahmi@gmail.com', 'Lelaki', 'Islam', 'Melayu', 'KTDI',  
    '23423', 'Pahang', '0198779670', '-', 3500.00, '0123', '1000'  
);
```

(b) Insert value into member application – Family Details

```
INSERT INTO member_applications (  
    Nama_pewaris, hubungan_pewaris, no_kp_pewaris  
) VALUES (  
    'Afiq', 'Abang', '0236987415'  
);
```

(c) Insert value into member application – Payment Details

```
INSERT INTO member_applications (  
    fee_masuk, receipt_path  
) VALUES (  
    50, 'receipt.png'  
);
```

4.2.3 Loan Application

DDL:

(a) Create loan application table

```
CREATE TABLE loan_application_details (  
    loan_id INT AUTO_INCREMENT PRIMARY KEY,  
    loan_type VARCHAR(255) NOT NULL,  
    loan_amount DECIMAL(10,2) NOT NULL,  
    loan_duration INT NOT NULL,  
    salary DECIMAL(10,2) NOT NULL,  
    loan_date DATE DEFAULT NULL,  
    first_guarantorname VARCHAR(255) DEFAULT NULL,  
    first_staffnum INT DEFAULT NULL,  
    first_guarantorIC VARCHAR(20) DEFAULT NULL,  
    sec_guarantorname VARCHAR(255) DEFAULT NULL,  
    sec_staffnum INT DEFAULT NULL,  
    sec_guarantorIC VARCHAR(20) DEFAULT NULL,  
    review_status ENUM('reviewed','not_reviewed') DEFAULT 'not_reviewed',  
    status ENUM('pending','accepted','rejected') DEFAULT 'pending',  
    bank_account_number VARCHAR(16) DEFAULT NULL,  
    Bank_account_name VARCHAR(255) DEFAULT NULL,  
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP  
);
```

DML:

(a) Insert Value for Deposit Saving Account inside the form (All Share Fee, Pay month)

```
INSERT INTO loan_application (  
  
    loan_type,    loan_amount,    loan_duration,    salary,    first_guarantorname,    first_staffnum,  
    first_guarantornum, sec_guarantorname, sec_staffnum, sec_guarantornum, bank_account_num, bank-  
    account_name)
```

```
VALUES ('Al-Bai', 7000.00, 12, 3000.00, 'Mohd Rashid', 1710, '0139876543', 'Abdullah Kadir', 1222,  
'0195567990', 1001, "4722 4833 5564 6544", "MAYBANK");
```

(b) Insert Value for Loan Transaction Status

```
UPDATE pending_loan  
SET status = 'approved'  
WHERE loan_id = 1;
```

4.2.4 Savings Application

DDL:

```
CREATE TABLE syer_accounts (  
  
    account_id INT PRIMARY KEY AUTO_INCREMENT,  
    member_num VARCHAR(4) NOT NULL,  
    modah_syer DECIMAL(10,2) DEFAULT 0.00,  
    modah_fee DECIMAL(10,2) DEFAULT 0.00,  
    fixed_saving DECIMAL(10,2) DEFAULT 0.00,  
    member_saving DECIMAL(10,2) DEFAULT 0.00,  
    fund_donation DECIMAL(10,2) DEFAULT 0.00,  
    payment_method VARCHAR(50),  
  
);  
  
CREATE TABLE pending_transaction (  
    transaction_id INT PRIMARY KEY AUTO_INCREMENT,  
    account_id INT PRIMARY KEY AUTO_INCREMENT,  
    member_num VARCHAR(4) NOT NULL,  
    transaction_type_id INT NOT NULL,  
    transaction_status_ ENUM('approved','rejected') DEFAULT NULL,  
    modah_syer_inc DECIMAL(10,2) DEFAULT 0.00,  
    modah_fee_inc DECIMAL(10,2) DEFAULT 0.00,  
    fixed_saving_inc DECIMAL(10,2) DEFAULT 0.00,  
    member_saving_inc DECIMAL(10,2) DEFAULT 0.00,  
    fund_donation_inc DECIMAL(10,2) DEFAULT 0.00,  
    transaction_date DATE NOT NULL,  
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,  
  
);
```

DML:**(a) Insert Value for Deposit Saving Account inside the form (All Share Fee, Pay month)**

```
UPDATE syer_accounts AS s
JOIN pending_savings AS p
  ON s.account_id = p.account_id
SET
  s.modah_syer = s.modah_syer + p.modah_syer_inc,
  s.modah_fee = s.modah_fee + p.modah_fee_inc,
  s.fixed_saving = s.fixed_saving + p.fixed_saving_inc,
  s.member_saving = s.member_saving + p.member_saving_inc,
  s.fund_donation = s.fund_donation + p.fund_donation_inc
WHERE p.transaction_id = 123
AND p.transaction_status_ = 'approved';
```

(b) Update the transaction status for the deposit

```
UPDATE pending_savings SET transaction_status_ = 'rejected',
WHERE transaction_id = 1;
```

4.2.5 Review Applications

DDL:

```
CREATE TABLE Admin (  
    user_id INT AUTO_INCREMENT PRIMARY KEY,  
    password VARCHAR(255) UNIQUE NOT NULL,  
    email VARCHAR(255) NOT NULL,  
    registration_date DATETIME DEFAULT CURRENT_TIMESTAMP  
);
```

```
CREATE TABLE Board_Member (  
    user_id INT AUTO_INCREMENT PRIMARY KEY,  
    password VARCHAR(255) UNIQUE NOT NULL,  
    email VARCHAR(255) NOT NULL,  
    registration_date DATETIME DEFAULT CURRENT_TIMESTAMP  
);
```

```
CREATE TABLE MemberApplicationDetails (  
    memberNum INT AUTO_INCREMENT PRIMARY KEY,  
    pfNum VARCHAR(20) UNIQUE NOT NULL,  
    ic_number VARCHAR(20) UNIQUE NOT NULL,  
    name VARCHAR(255) NOT NULL,  
    marriage_status ENUM('Single', 'Married', 'Divorced', 'Widowed') NOT NULL,  
    home_address VARCHAR(255) NOT NULL,  
    office_address VARCHAR(255),  
    phoneNum VARCHAR(20) UNIQUE NOT NULL,  
    gender VARCHAR(50) NOT NULL,  
    religion VARCHAR(50) NOT NULL,
```



```

race VARCHAR(50) NOT NULL,
review_status ENUM('reviewed', 'not reviewed') DEFAULT 'not reviewed',
status ENUM('accepted', 'rejected', 'pending') DEFAULT 'pending'
);

```

```

CREATE TABLE LoanApplicationDetails (
    loan_id INT AUTO_INCREMENT PRIMARY KEY,
    loan_type VARCHAR(50) NOT NULL,
    loan_amount DECIMAL(10,2) NOT NULL,
    loan_duration INT(10) NOT NULL,
    monthly_repayment DECIMAL(10,2) NOT NULL,
    salary DECIMAL(10,2) NOT NULL,
    loan_date DATE NOT NULL,
    first_guarantorname VARCHAR(255) NOT NULL,
    first_staffID VARCHAR(50) NOT NULL,
    first_guarantornum VARCHAR(20) NOT NULL,
    sec_guarantorname VARCHAR(255) NOT NULL,
    sec_staffID VARCHAR(50) NOT NULL,
    sec_guarantornum VARCHAR(20) NOT NULL,
    memberNum INT NOT NULL,
    review_status ENUM('Pending', 'Approved', 'Rejected') DEFAULT 'Pending',
    status ENUM('Active', 'Inactive') DEFAULT 'Active',
    FOREIGN KEY (memberNum) REFERENCES MemberApplicationDetails(memberNum) ON
    DELETE CASCADE
);

```

DML:

(a) Query 'not reviewed' applications in admin view

```
SELECT * FROM member_application WHERE review_status = 'not reviewed';
```

```
SELECT * FROM loan_applications WHERE review_status = 'not reviewed';
```

(b) Query 'pending' applications in ALK view

```
SELECT * FROM member_application WHERE status = 'reviewed' AND status =  
'pending';
```

```
SELECT * FROM loan_applications WHERE status = 'reviewed' AND status = 'pending';
```

(c) Log in into admin dashboard

```
INSERT INTO Admin (email, password) VALUES ('ali_04@gmail.com', 'admin123');
```

(d) Log into ALK dashboard

```
INSERT INTO Board_Member (email, password) VALUES ('afiq_06@gmail.com',  
'alk123');
```

4.2.6 Loan Calculator

DDL:

```
CREATE TABLE Member (  
    memberID INT AUTO_INCREMENT PRIMARY KEY,  
    password VARCHAR(255) NOT NULL,  
    sharesTotal DECIMAL(10,2) NOT NULL,  
    startDate DATE NOT NULL,  
    endDate DATE  
);  
  
CREATE TABLE LoanCalculator (  
    calculatorID INT AUTO_INCREMENT PRIMARY KEY,  
    memberID INT NOT NULL,  
    loanAmount DECIMAL(10,2) NOT NULL,  
    loanDuration INT NOT NULL,  
    loanType VARCHAR(50) NOT NULL,  
    interestRate DECIMAL(5,2) NOT NULL,  
    FOREIGN KEY (memberID) REFERENCES Member(memberID) ON DELETE CASCADE  
);  
  
CREATE TABLE LoanRate (  
    loan_id INT AUTO_INCREMENT PRIMARY KEY,  
    loan_type VARCHAR(50) NOT NULL,  
    interestRate DECIMAL(5,2) NOT NULL  
);  
  
CREATE TABLE LoanCalc_History (  
    historyID INT AUTO_INCREMENT PRIMARY KEY,  
    memberID INT NOT NULL,  
    loanAmount DECIMAL(10,2) NOT NULL,  
    loanDuration INT NOT NULL,  
    loanType VARCHAR(50) NOT NULL,  
    interestRate DECIMAL(5,2) NOT NULL,  
    FOREIGN KEY (memberID) REFERENCES Member(memberID) ON DELETE CASCADE  
);
```

DML:

(a) Members fill up the fields in loan calculator

```
INSERT INTO LoanCalculator (loanAmount, loanDuration, loanType)
```

```
VALUES
```

```
(50000.00, 24, 'Al-Bai'),
```

```
(200000.00, 60, 'Al-Innah'),
```

```
(15000.00, 12, 'Vehicle Restoration');
```

(b) Loan calculator query the interest rate based on loan type

```
SELECT interestRate FROM LoanRate WHERE loan_type = 'Al-Bai';
```

```
SELECT interestRate FROM LoanRate WHERE loan_type = 'Al-Innah';
```

```
SELECT interestRate FROM LoanRate WHERE loan_type = 'Vehicle Restoration';
```

4.2.7 Admin Report

DDL:

```
CREATE TABLE Member (  
  memberNum INT PRIMARY KEY,  
  startDate DATE NOT NULL,  
  endDate DATE NOT NULL );
```

```
CREATE TABLE Member_Savings (  
  Savings_id INT PRIMARY KEY,  
  memberNum INT NOT NULL,  
  month DATE NOT NULL,  
  amount DECIMAL(10, 2) NOT NULL,  
  FOREIGN KEY (memberNum) REFERENCES Member(memberNum) );
```

```
CREATE TABLE Shares (  
  share_id INT PRIMARY KEY,  
  memberNum INT NOT NULL,  
  total_shares DECIMAL(10, 2) NOT NULL,  
  FOREIGN KEY (memberNum) REFERENCES Member(memberNum) );
```

```
CREATE TABLE Dividend (  
  dividend_id INT PRIMARY KEY,  
  year INT NOT NULL,  
  rate DECIMAL(10, 2) NOT NULL );
```

```
CREATE TABLE Reports (  
  report_id INT PRIMARY KEY,  
  memberNum INT NOT NULL,
```

```
report_date DATE NOT NULL,  
monthly_savings DECIMAL(10, 2) NOT NULL,  
yearly_profit DECIMAL(10, 2) NOT NULL,  
FOREIGN KEY (memberNum) REFERENCES Member(memberNum) );
```

DML:

```
INSERT INTO Member (memberNum, startDate, endDate)  
VALUES (1, '2023-01-01', '2024-01-01');
```

```
INSERT INTO Member_Savings (Savings_id, memberNum, month, amount)  
VALUES (101, 1, '2023-12-01', 1000.00);
```

Calculate Yearly Profit for a Member

```
SELECT memberNum, SUM(amount) AS yearlyProfit  
FROM Member_Savings  
WHERE memberNum=1;
```

```
UPDATE Member  
SET endDate = '2025-01-01'  
WHERE memberNum = 1;
```

```
DELETE FROM Member  
WHERE memberNum = 1;
```

5. Summary

In this phase, we modified the very beginning Global Conceptual ERD to a clearer Logical ERD for the KADA Cooperative Online System. We also demonstrated how various elements, such as Member, LoanApplicationDetails, and Syer_Account, were associated with one another and merged similar concepts afterward. In this manner, each table revealed the joining rules, the lending process, and the shares handling.

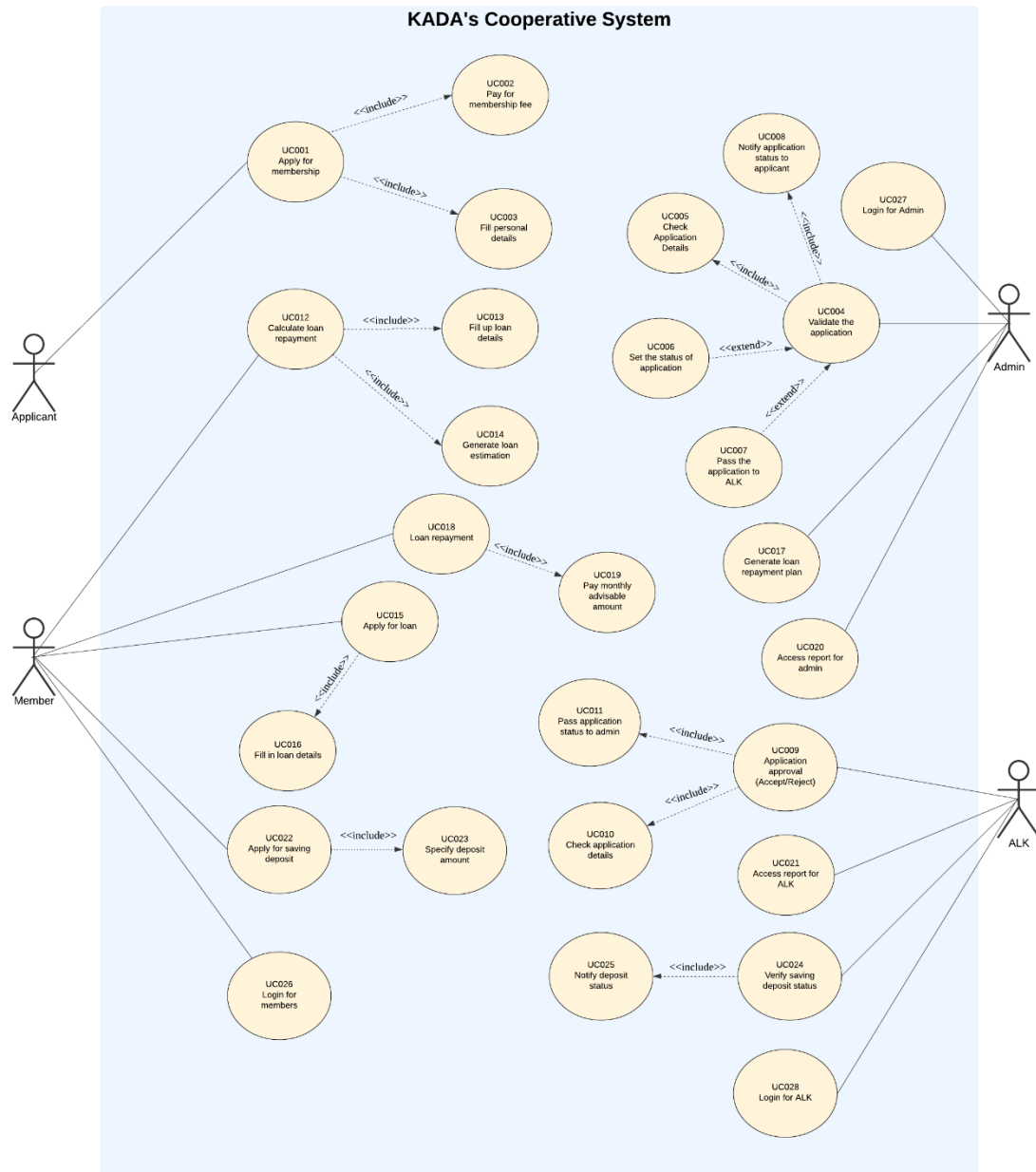
We made some essential elements like Member, Admin, Board_Member, and Fee into actual tables in the database. These tables possess unique features (such as primary and foreign keys) for uniquely identifying the records and establishing the right data relationships. We applied normalization techniques (up to the level of BCNF) for eliminating duplicate data and data redundancy. This is essential for keeping the data about members, loans, and savings up-to-date for the activities of the cooperative.

We got enhanced by including data dictionary detailing each table's attributes, data type, and business rule. This dictionary is useful for stakeholders and programmers alike to grasp the structure of tables such as LoanApplicationDetails, MemberApplicationDetails, and Shares. This ensures all the changes made in the future will cater to the needs of the system. In demonstrating the relationships and the dependences, you establish the solid foundation for secure transactions, data retrievals, and maintainability.

The Logical ERD is now a solid blueprint for the connection between membership, loans, and investments in shares within the system. This is building a solid structure for the database, and this benefits the KADA Cooperative by minimizing paperwork, being easier to expand, and providing a clearer history for all activities by the members. Overall, this piece of work benefits the applicants, members, and admins by providing them with easier access, and the system can evolve for the future.

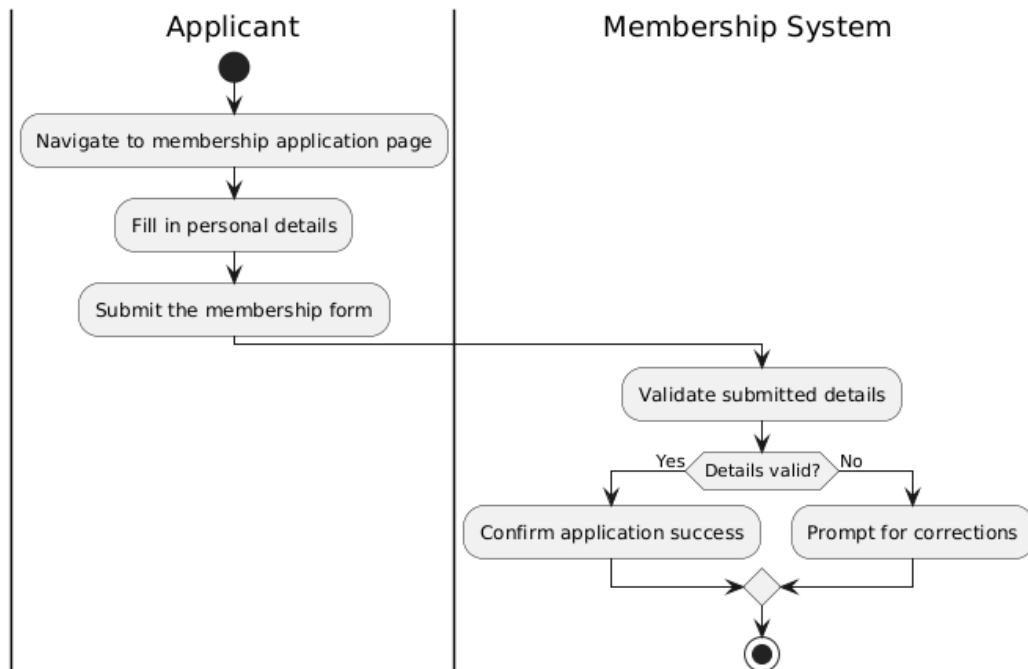
6. Appendices

6.1. Use Case Diagram for the system

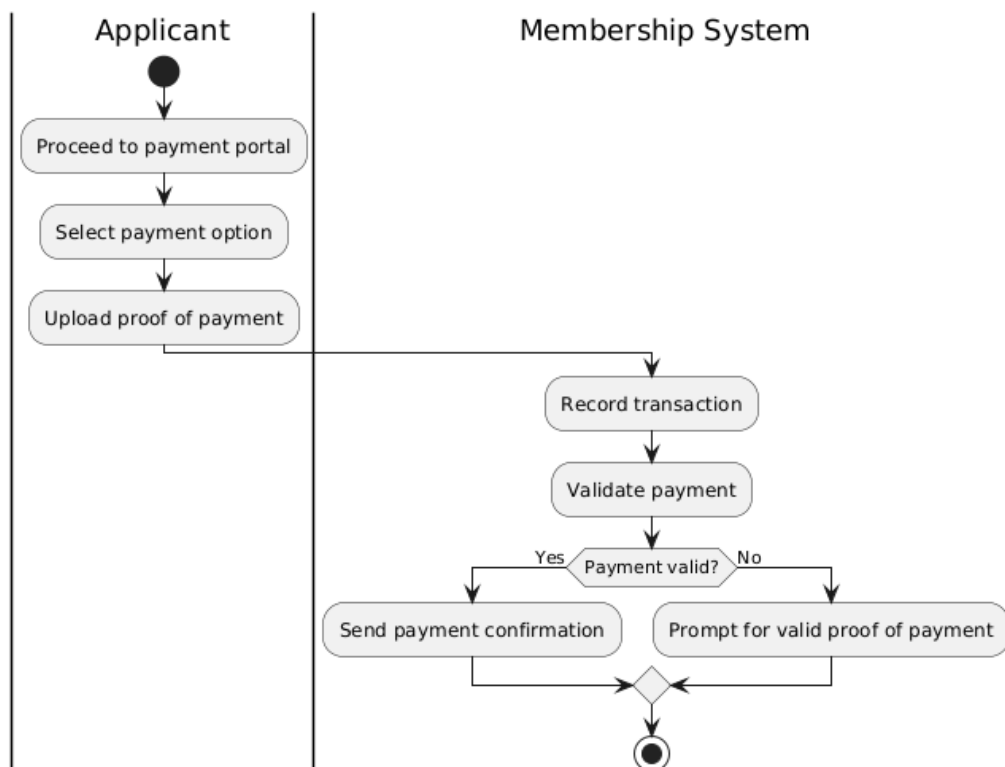


6.2. Activity Diagrams for every use case identified

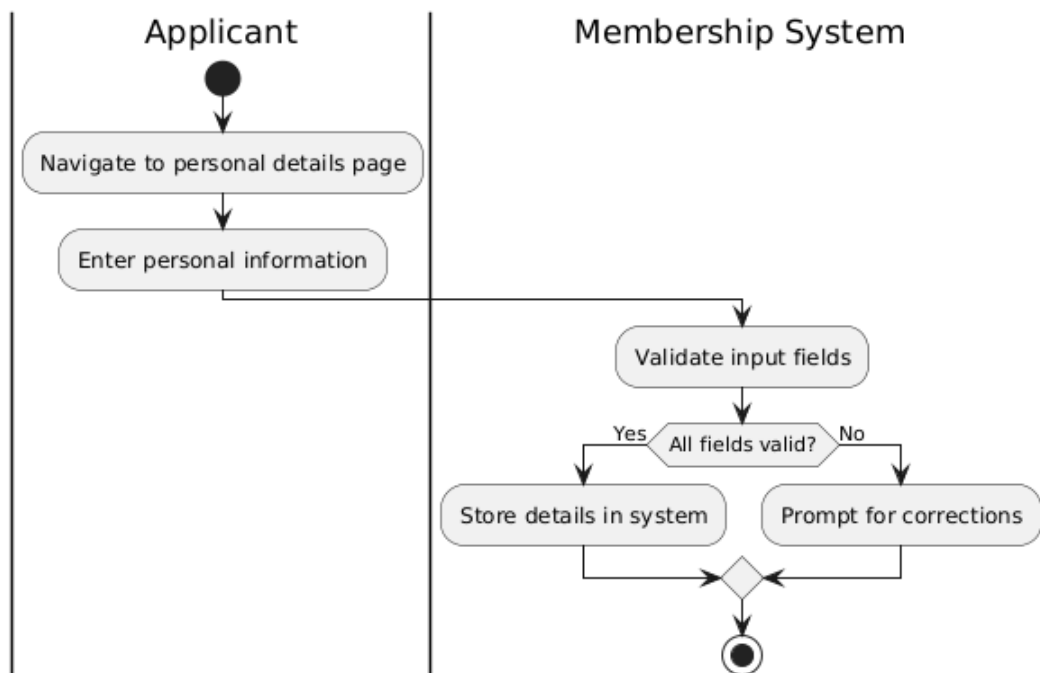
6.2.1 Activity Diagram 1: UC001



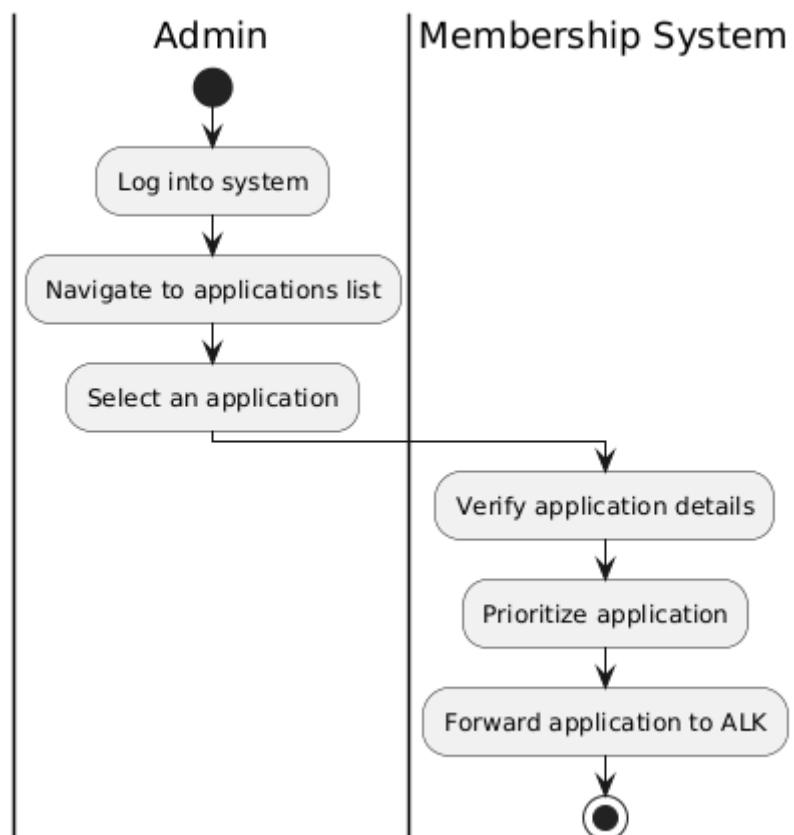
6.2.2 Activity Diagram 2: UC002



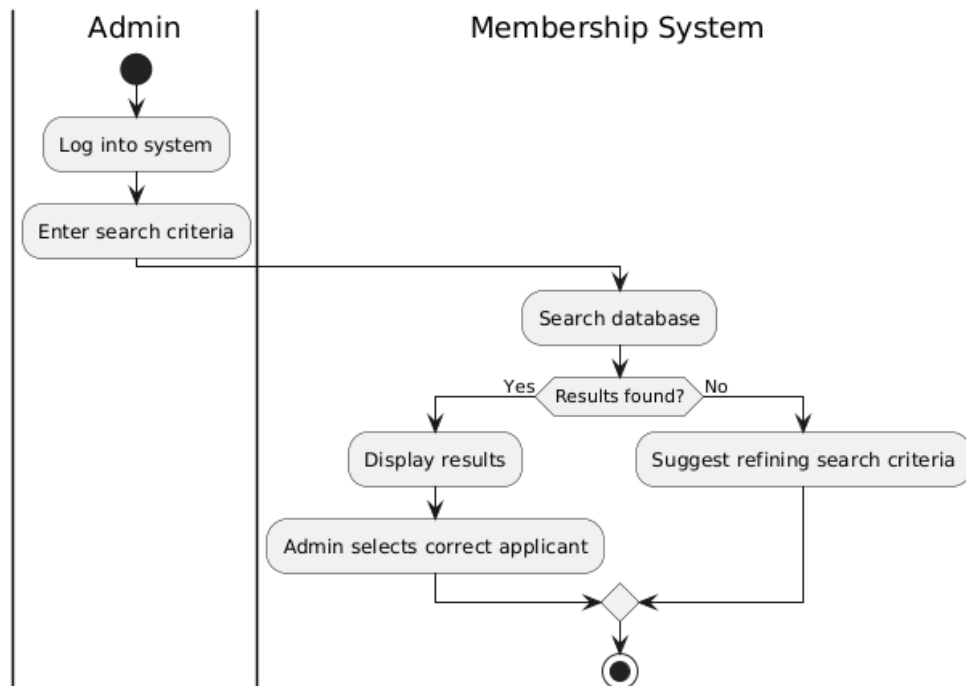
6.2.3 Activity Diagram 3: UC003



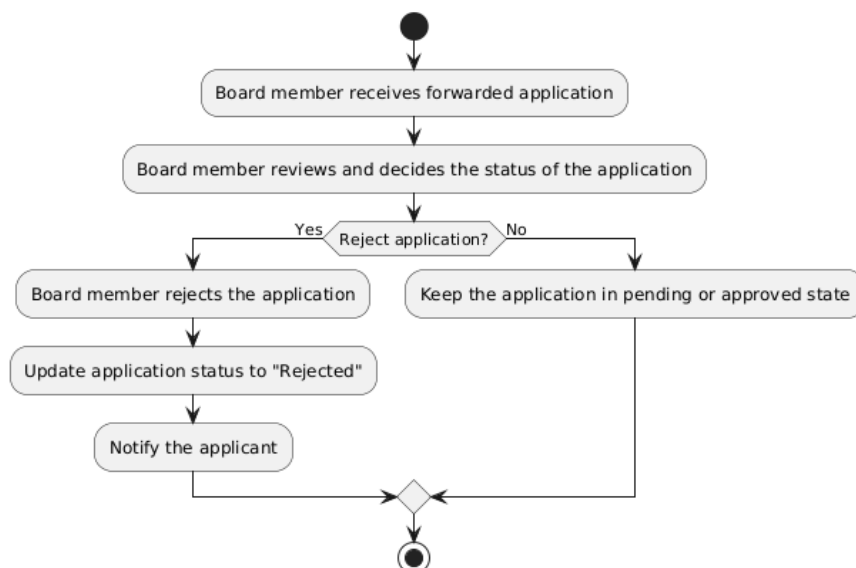
6.2.4 Activity Diagram 4: UC004



6.2.5 Activity Diagram 5: UC005

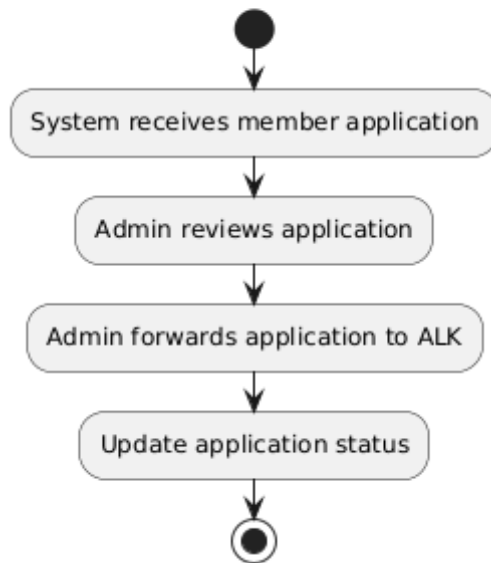


6.2.6 Activity Diagram 6: UC006



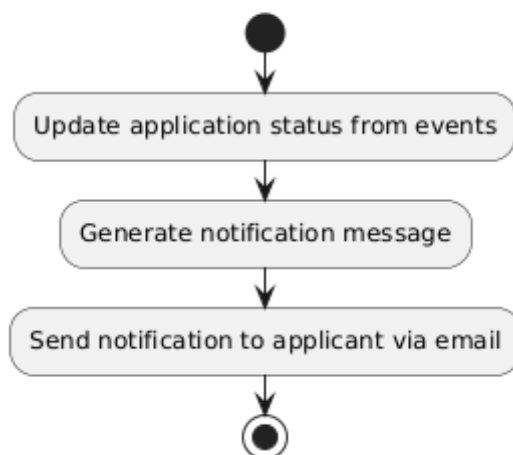
6.2.7 Activity Diagram 7: UC007

Passing Application to ALK



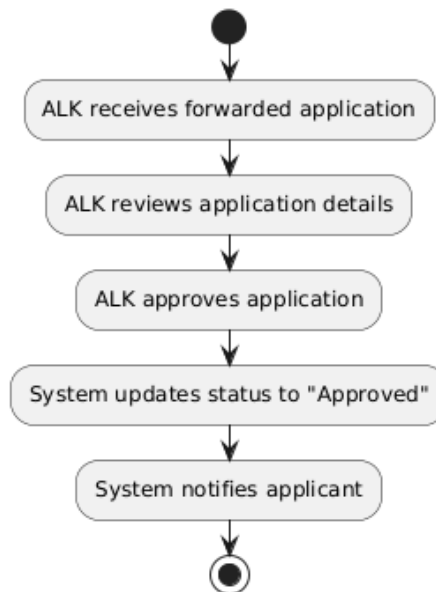
6.2.8 Activity Diagram 8: UC008

Notifying Application Status



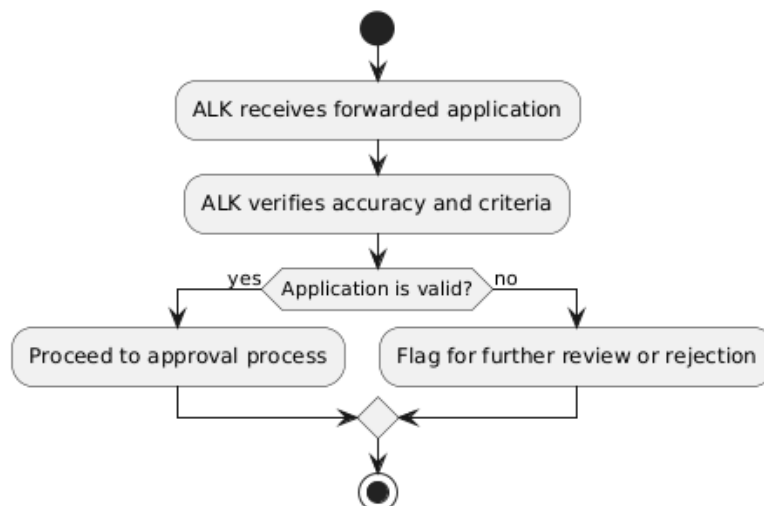
6.2.9 Activity Diagram 9: UC009

Application Approval Process

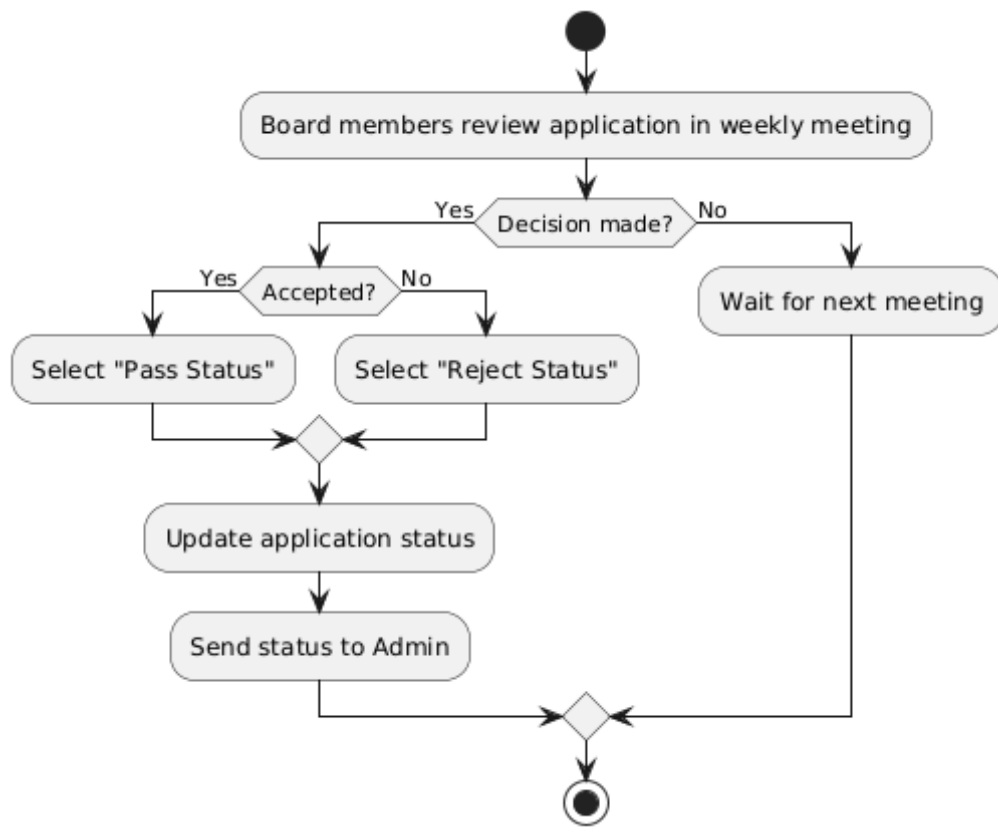


6.2.10 Activity Diagram 10: UC010

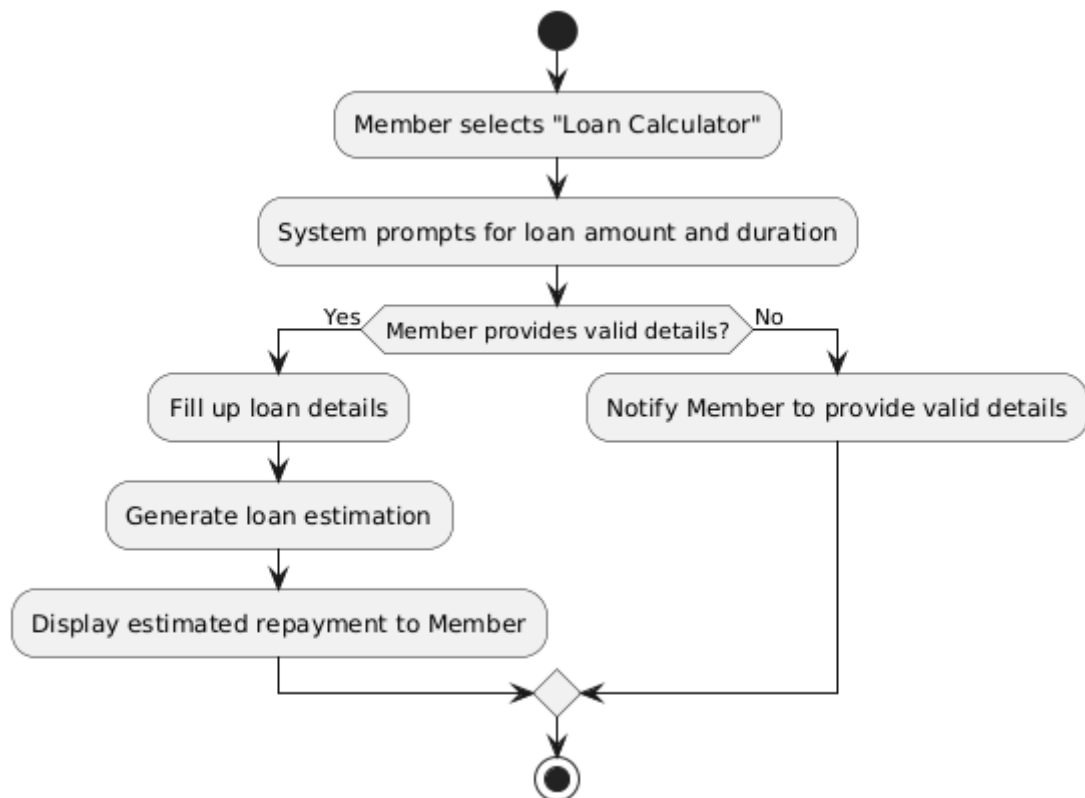
Application Validation Process



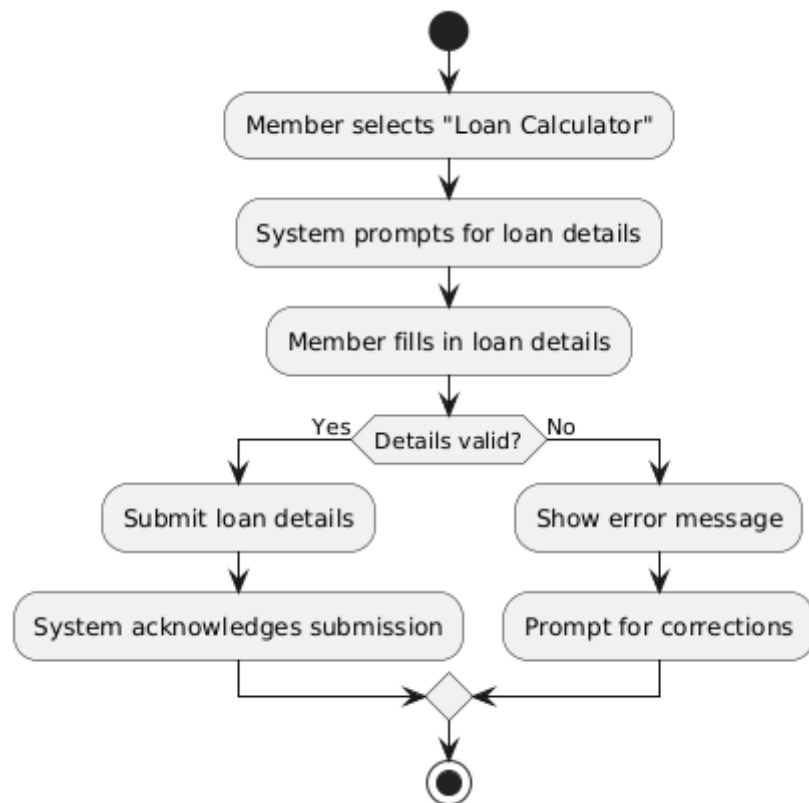
6.2.11 Activity Diagram 11: UC011



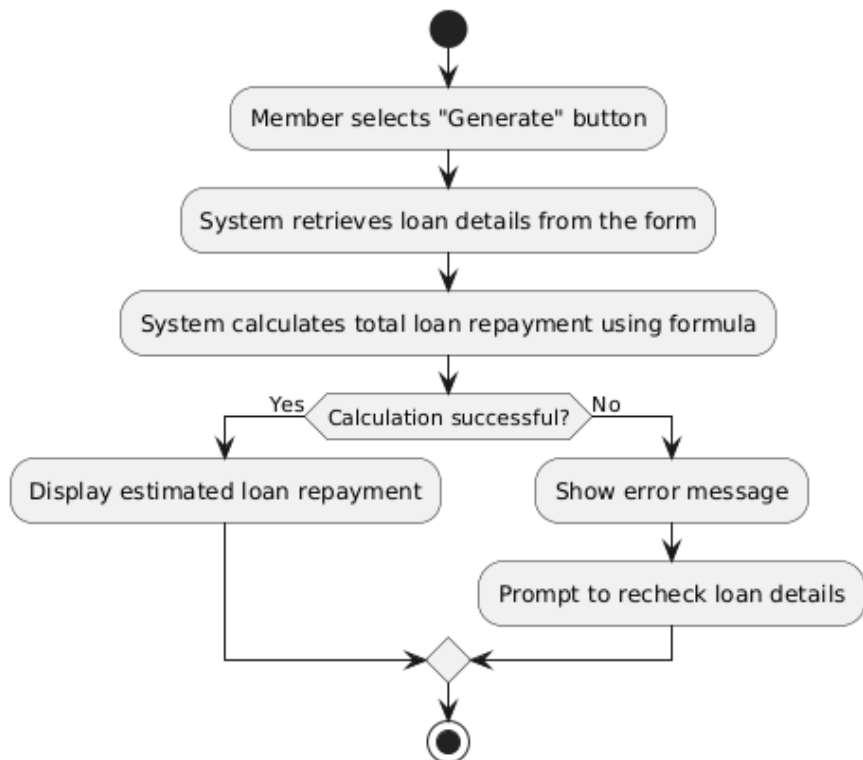
6.2.12 Activity Diagram 12: UC012



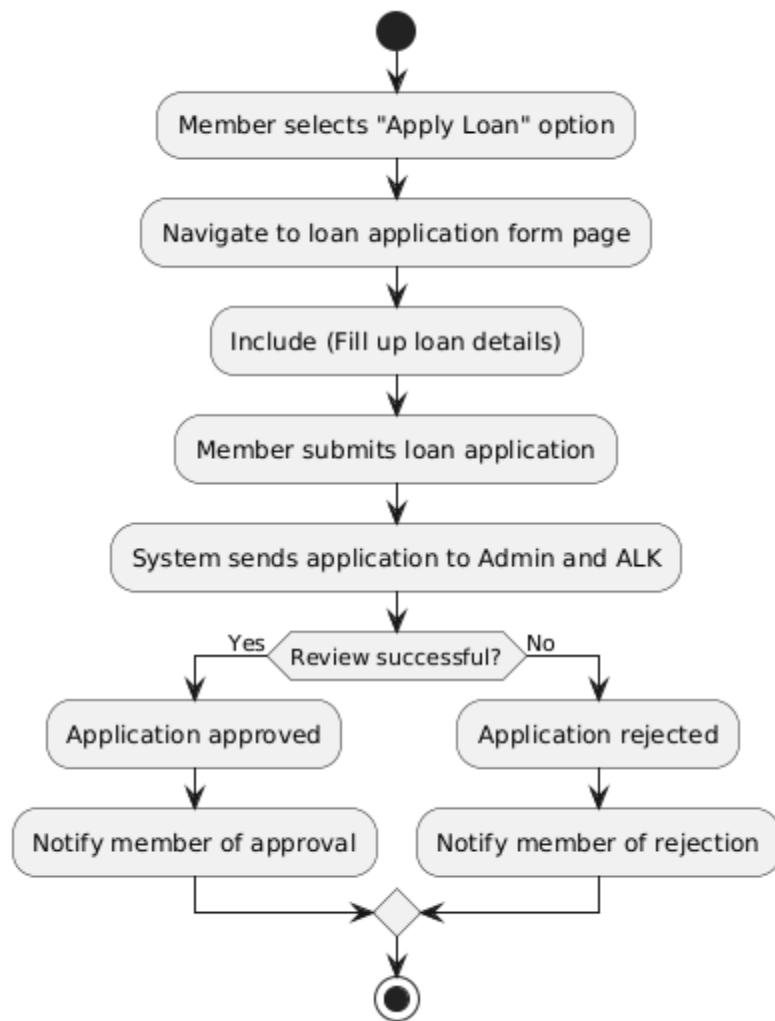
6.2.13 Activity Diagram 13: UC013



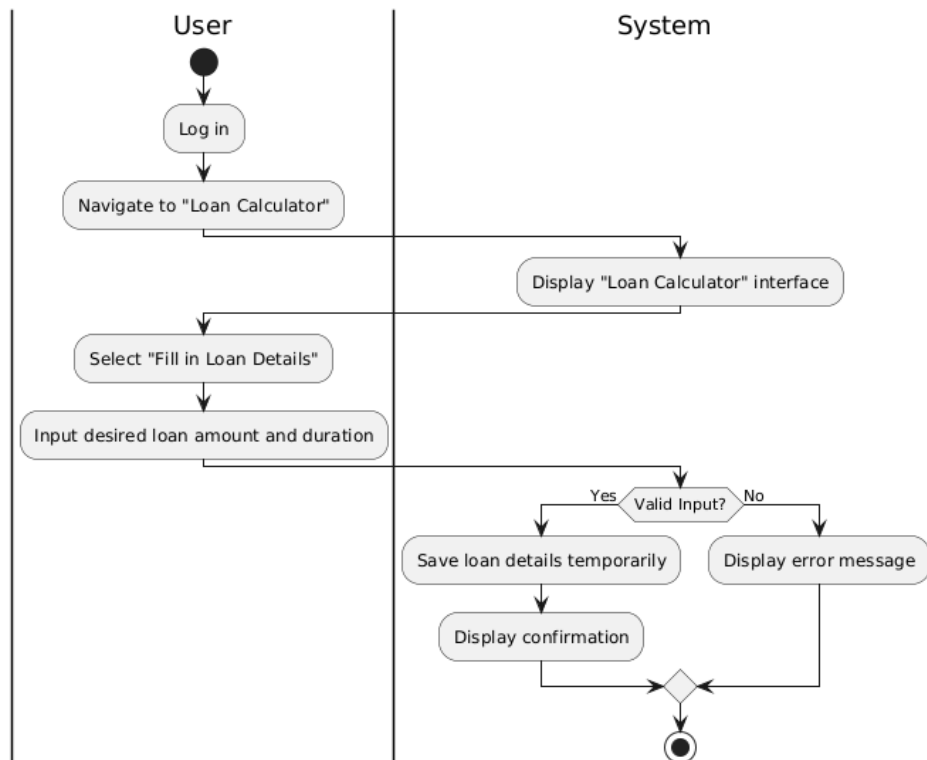
6.2.14 Activity Diagram 14: UC014



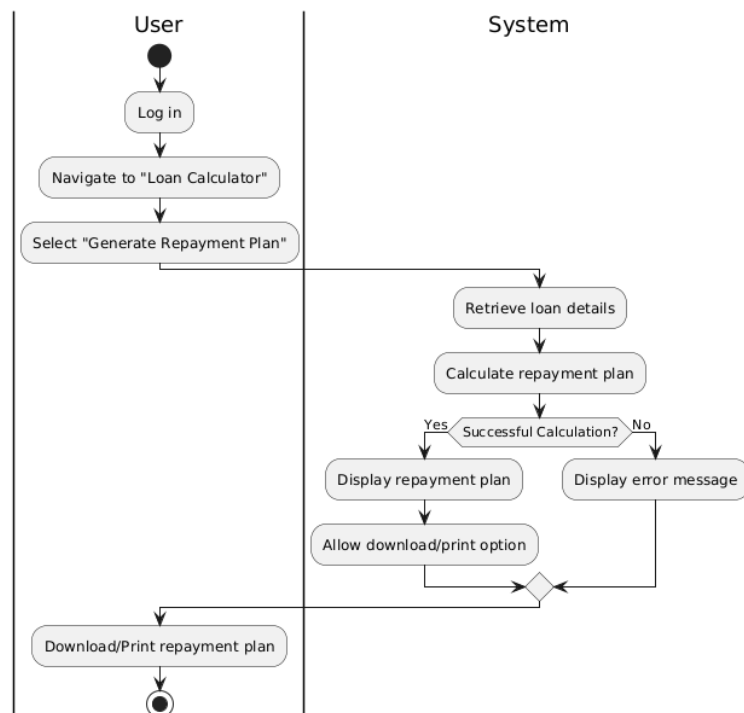
6.2.15 Activity Diagram 15: UC015



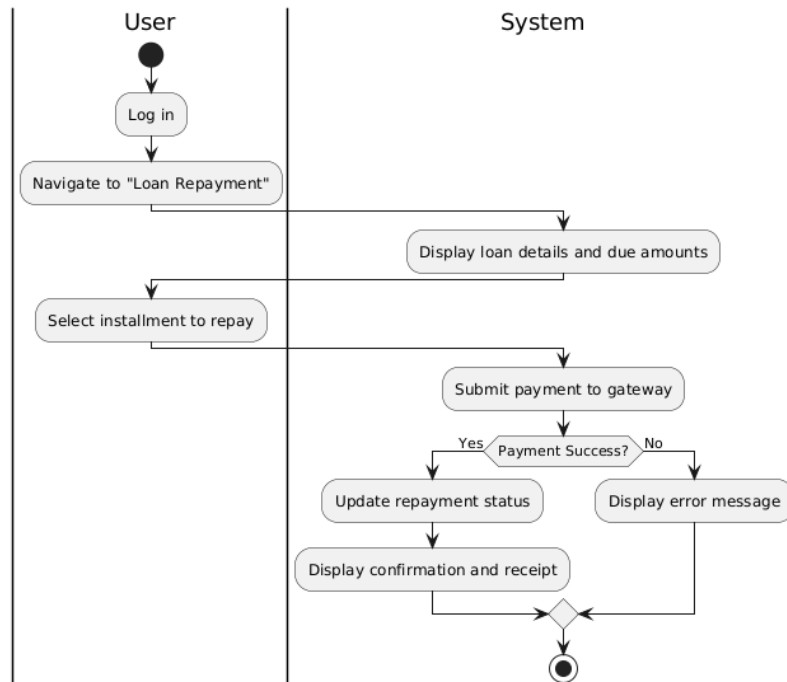
6.2.16 Activity Diagram 16: UC016



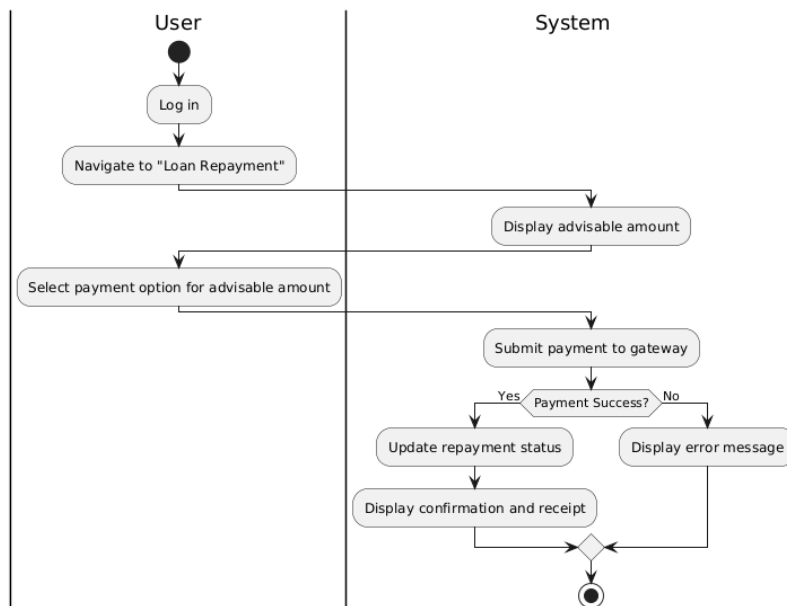
6.2.17 Activity Diagram 17: UC017



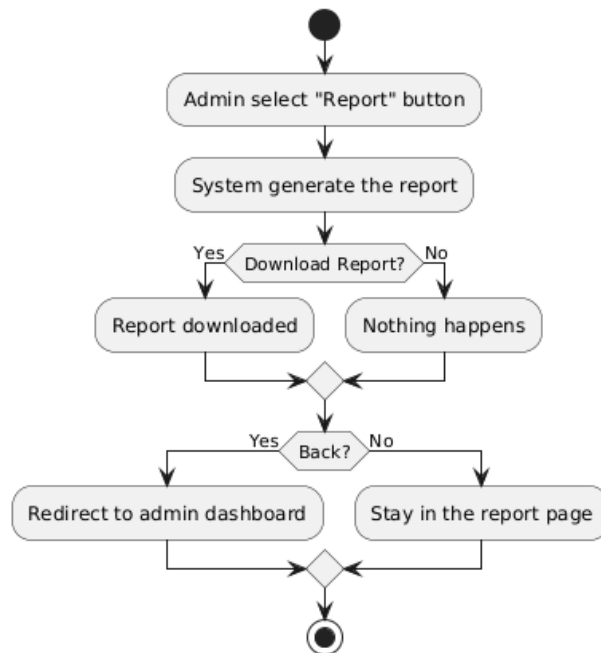
6.2.18 Activity Diagram 18: UC018



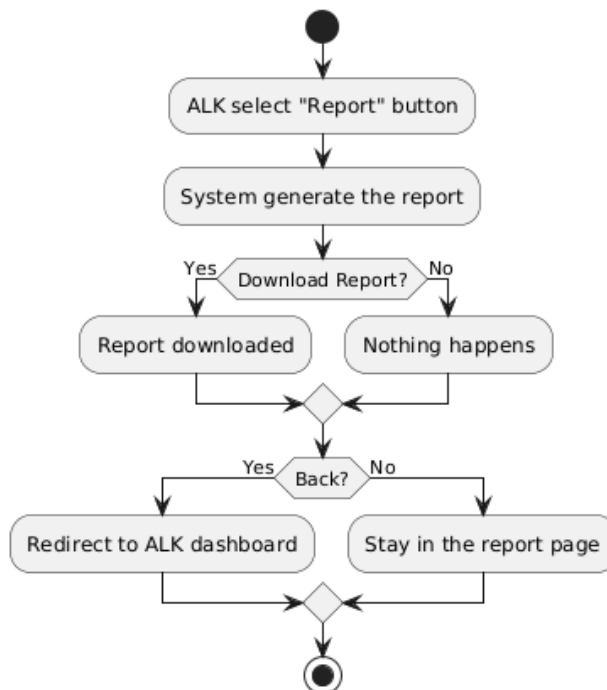
6.2.19 Activity Diagram 19: UC019



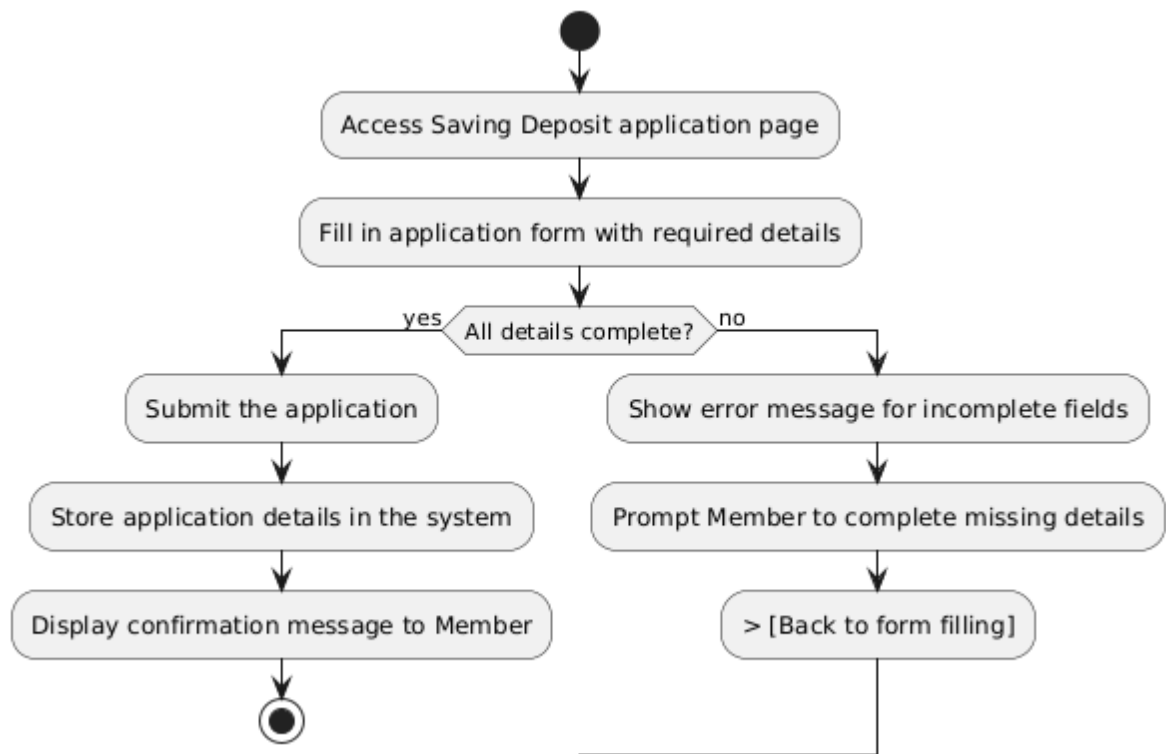
6.2.20 Activity Diagram 20: UC020



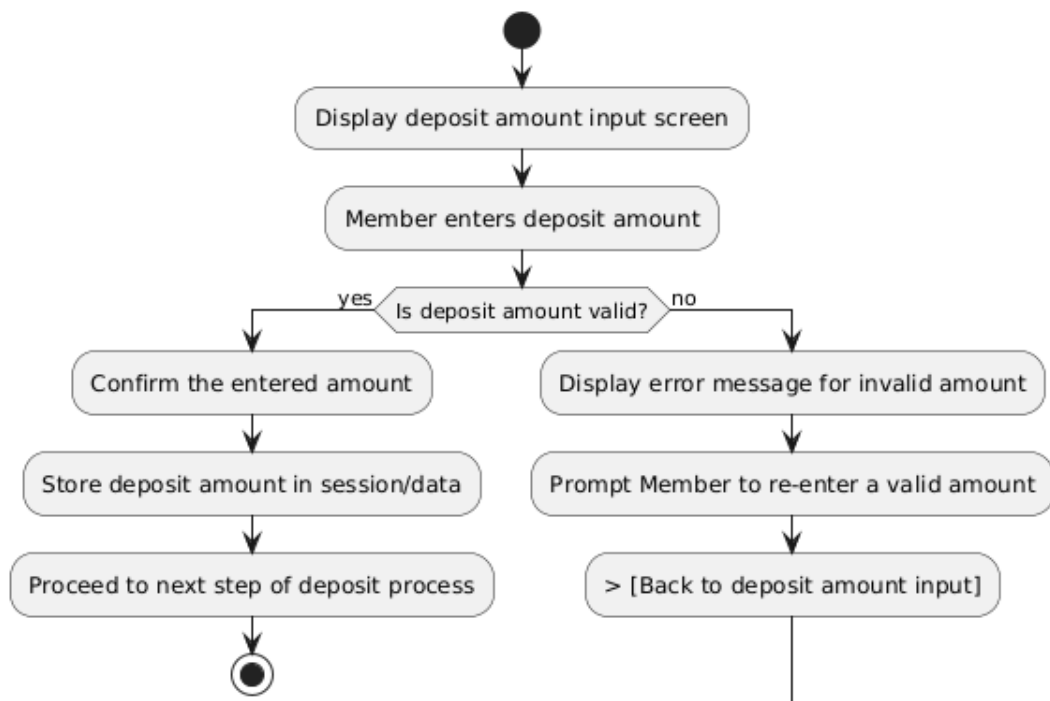
6.2.21 Activity Diagram 21: UC021



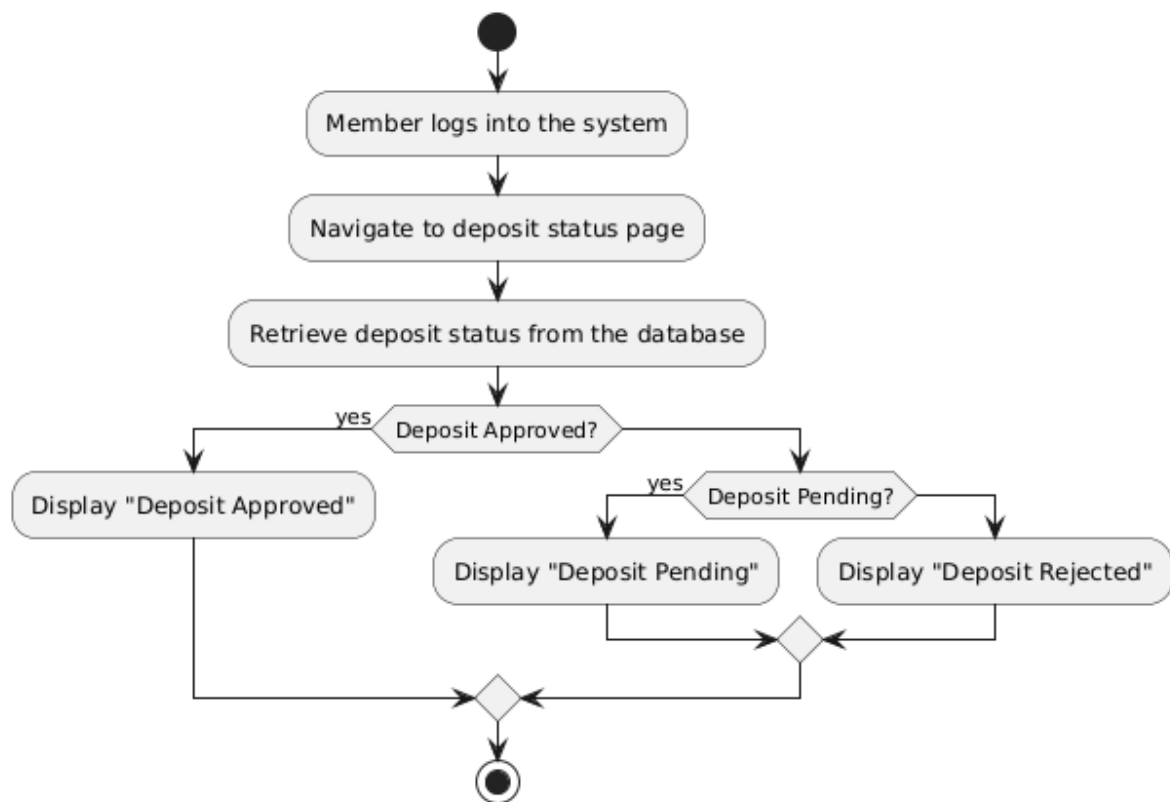
6.2.22 Activity Diagram 22: UC022



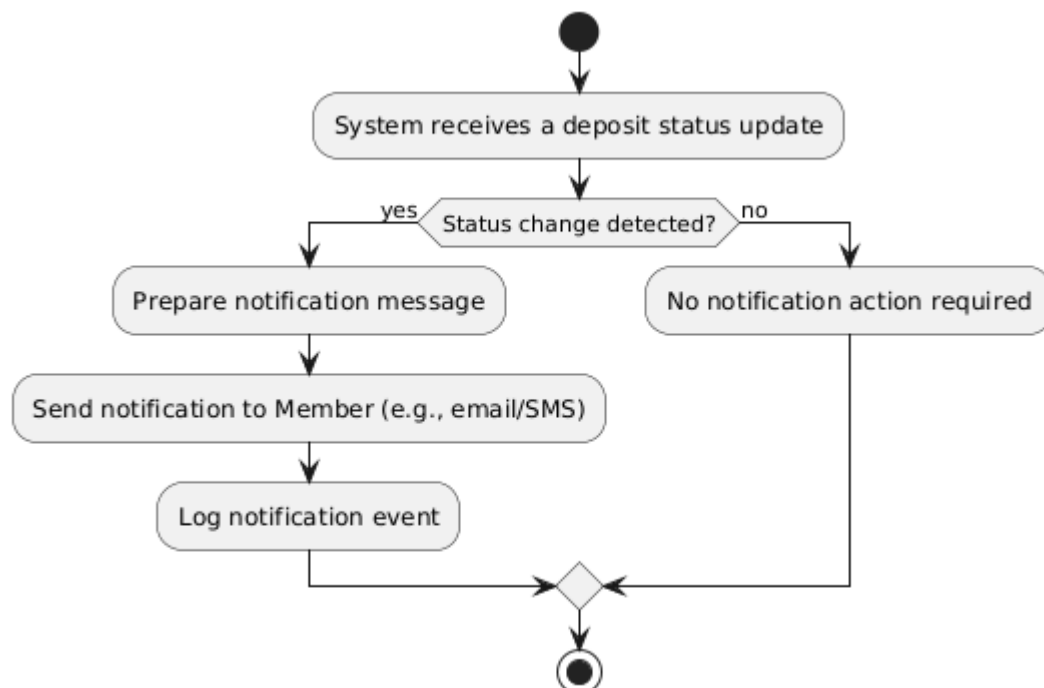
6.2.23 Activity Diagram 23: UC023



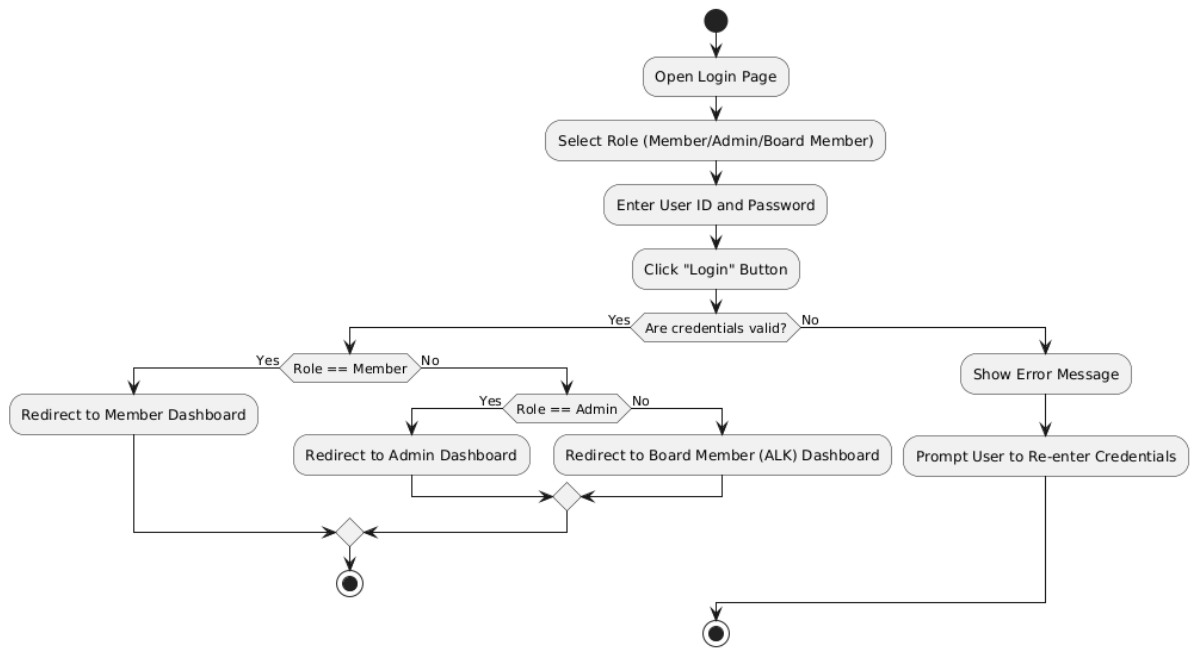
6.2.24 Activity Diagram 24: UC024



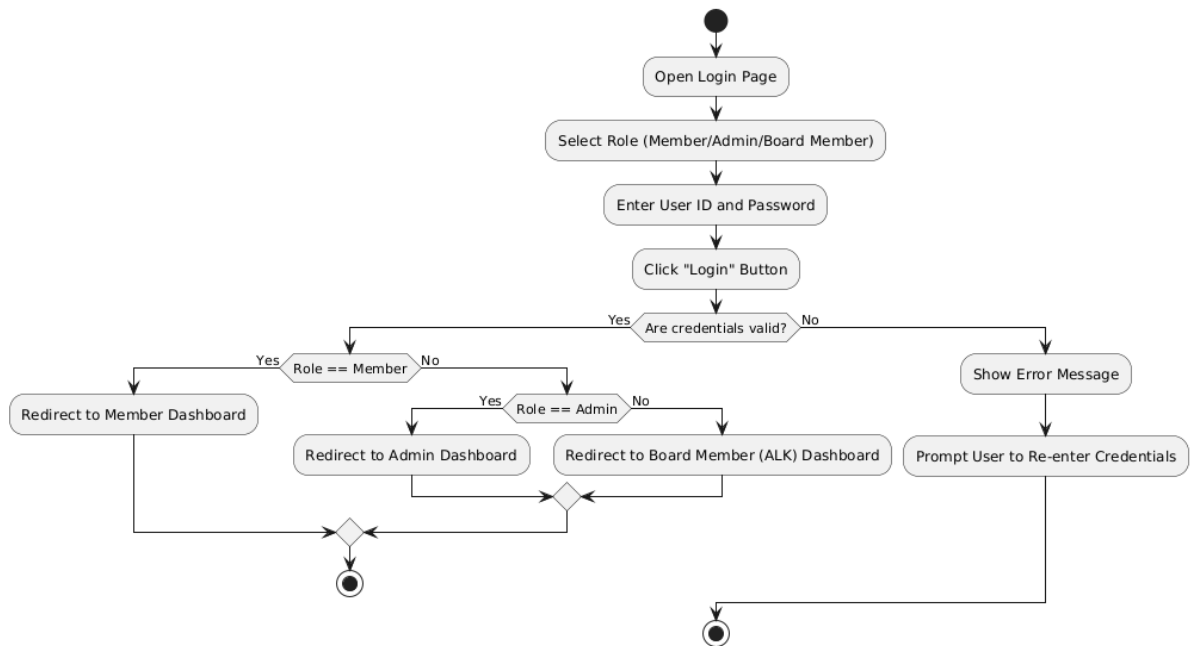
6.2.25 Activity Diagram 25: UC025



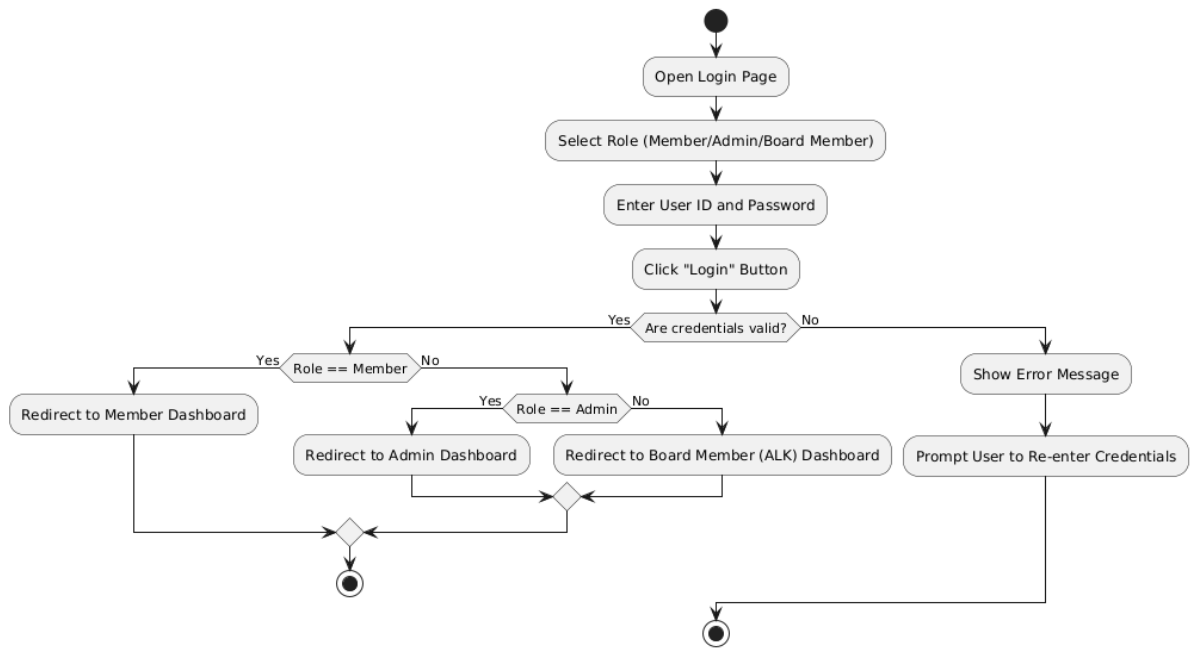
6.2.26 Activity Diagram 26: UC026



6.2.27 Activity Diagram 27: UC027



6.2.28 Activity Diagram 28: UC028



6.3. Meeting Log